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CHAPTER



VECTORS

Vectors:

A vector quantity is that possesses both magnitude and direction i.e. displacement, velocity, weight, force etc.

Scalar:

A scalar quantity is that possesses only magnitude. It can be specified by a number i.e. mass, time, density, length, volume etc.

Magnitude/Length/Norm/Modulus of a Vector:

The positive real number, which is measure of the length of the vector, is called modulus, length, magnitude or norm of a vector.

Formula $\hat{v} = \frac{v}{|v|}$

Zero Vector:

If terminal point B of a vector $\vec{AB}$ coincides with its initial point A, then $|\vec{AB}| = 0$ called zero vector or Null vector.

Position vector:

The vector, whose initial point O is origin & whose terminal point is P, is called position vector of OP.


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EXERCISE 7.1

Q.1 Write the vector $\vec{PQ}$ in the form $xi + yj$.

(i) $P(2, 3), Q(6, -2)$

Solution:

$P(2, 3), Q(6, -2)$

$$\begin{aligned}\vec{PQ} &= \vec{OQ} - \vec{OP} \text{ -- position vector of P} \\ &= (6 - 2)\underline{i} + (-2 - 3)\underline{j} = 4\underline{i} - 5\underline{j}\end{aligned}$$

(ii) $P(0, 5), Q(-1, -6)$

Solution:

$$\begin{aligned}\vec{PQ} &= \vec{OQ} - \vec{OP} \\ &= (-1 - 0)\underline{i} + (-6 - 5)\underline{j} = -\underline{i} - 11\underline{j}\end{aligned}$$



Q.2: Find the magnitude of the vector $\underline{u}$.

Formula

Magnitude or length or Norm of $\underline{v} = x\underline{i} + y\underline{j} + z\underline{k}$ is $|\underline{V}| = \sqrt{x^2 + y^2 + z^2}$

(i) $\underline{u} = 2\underline{i} - 7\underline{j}$

Solution:

$$\underline{u} = 2\underline{i} - 7\underline{j}$$

$$|\underline{u}| = \sqrt{(2)^2 + (-7)^2} = \sqrt{4 + 49} = \sqrt{53}$$

(ii) $\underline{u} = \underline{i} + \underline{j}$

Solution:

$$\underline{u} = \underline{i} + \underline{j}$$

$$|\underline{u}| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$$

(ii) $\underline{u} = [3, -4]$ (Lahore Board 2005)

Solution:

$$\underline{u} = 3\underline{i} - 4\underline{j}$$

$$|\underline{u}| = \sqrt{(3)^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Q.3 If $\underline{u} = 2\underline{i} - 7\underline{j}$, $\underline{v} = \underline{i} - 6\underline{j}$ & $\underline{w} = -\underline{i} + \underline{j}$, find the following vectors.

(i) $\underline{u} + \underline{v} - \underline{w}$

Solution:

$$\begin{aligned} \underline{u} + \underline{v} - \underline{w} &= (2\underline{i} - 7\underline{j}) + (\underline{i} - 6\underline{j}) - (-\underline{i} + \underline{j}) \\ &= 2\underline{i} - 7\underline{j} + \underline{i} - 6\underline{j} + \underline{i} - \underline{j} = 4\underline{i} - 14\underline{j} \end{aligned} \quad \text{Ans.}$$

(ii) $2\underline{u} - 3\underline{v} + 4\underline{w}$

Solution:

$$\begin{aligned} 2\underline{u} - 3\underline{v} + 4\underline{w} &= 2(2\underline{i} - 7\underline{j}) - 3(\underline{i} - 6\underline{j}) + 4(-\underline{i} + \underline{j}) \\ &= 4\underline{i} - 14\underline{j} - 3\underline{i} + 18\underline{j} - 4\underline{i} + 4\underline{j} = -3\underline{i} + 8\underline{j} \end{aligned}$$

(iii) $\frac{1}{2}\underline{u} + \frac{1}{2}\underline{v} + \frac{1}{2}\underline{w}$

Solution:

$$\begin{aligned} &= \frac{1}{2} [\underline{u} + \underline{v} + \underline{w}] \\ &= \frac{1}{2} [2\underline{i} - 7\underline{j} + \underline{i} - 6\underline{j} - \underline{i} + \underline{j}] \\ &= \frac{1}{2} [2\underline{i} - 12\underline{j}] \end{aligned}$$



$$= \frac{2}{2} [\underline{i} - 6\underline{j}] = \underline{i} - 6\underline{j}$$

Q.4 Find the sum of the vectors $\vec{AB}$ & $\vec{CD}$, given the four points A(1, -1), B (2, 0), C(-1, 3) & D (-2, 2)

Solution:

$$\begin{aligned}\vec{AB} &= \vec{OB} - \vec{OA} \\ &= (2-1)\underline{i} + (0+1)\underline{j} = \underline{i} + \underline{j}\end{aligned}$$

$$\begin{aligned}\vec{CD} &= \vec{OD} - \vec{OC} \\ &= (-2+1)\underline{i} + (2-3)\underline{j} = -\underline{i} - \underline{j}\end{aligned}$$

$$\text{Sum} = \vec{AB} + \vec{CD} = \underline{i} + \underline{j} - \underline{i} - \underline{j} = 0\underline{i} + 0\underline{j} = \text{Null vector}$$

Q.5 Find the vector from the point A to the origin, where $\vec{AB} = 4\underline{i} - 2\underline{j}$ and B is the point (-2, 5).

Solution:

$$\vec{AB} = 4\underline{i} - 2\underline{j}$$

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$\vec{AB} - \vec{OB} = -\vec{OA}$$

$$\vec{AB} - \vec{OB} = \vec{AO} \quad \because \vec{AO} = -\vec{OA}$$

$$\vec{AO} = (4\underline{i} - 2\underline{j}) - (-2\underline{i} + 5\underline{j})$$

$$\vec{AO} = 6\underline{i} - 7\underline{j}$$

Q.6 Find a unit vector in the direction of the vector given below

(i) $\underline{v} = 2\underline{i} - \underline{j}$ (Lahore Board 2009, 2010)

Solution:

$$\underline{v} = 2\underline{i} - \underline{j}$$

$$|\underline{v}| = \sqrt{(2)^2 + (-1)^2}$$

$$|\underline{v}| = \sqrt{4+1} = \sqrt{5}$$



Required unit vector is $\hat{v} = \frac{\underline{v}}{|\underline{v}|} = \frac{2\underline{i} - \underline{j}}{\sqrt{5}} = \frac{2}{\sqrt{5}} \underline{i} - \frac{1}{\sqrt{5}} \underline{j}$

$$(ii) \quad \underline{v} = \frac{1}{2} \underline{i} + \frac{\sqrt{3}}{2} \underline{j}$$

Solution:

$$\underline{v} = \frac{1}{2} \underline{i} + \frac{\sqrt{3}}{2} \underline{j}$$

$$|\underline{v}| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{3}{4}} = \sqrt{\frac{4}{4}} = 1$$

Required unit vector is $\hat{v} = \frac{\underline{v}}{|\underline{v}|} = \frac{\frac{1}{2} \underline{i} + \frac{\sqrt{3}}{2} \underline{j}}{1} = \frac{1}{2} \underline{i} + \frac{\sqrt{3}}{2} \underline{j}$ Ans.

$$(iii) \quad \underline{v} = \frac{-\sqrt{3}}{2} \underline{i} - \frac{1}{2} \underline{j}$$

Solution:

$$\underline{v} = \frac{-\sqrt{3}}{2} \underline{i} - \frac{1}{2} \underline{j}$$

$$|\underline{v}| = \sqrt{\left(\frac{-\sqrt{3}}{2}\right)^2 + \left(\frac{-1}{2}\right)^2} = \sqrt{\frac{3}{4} + \frac{1}{4}} = 1$$

Required unit vector $\hat{v} = \frac{\underline{v}}{|\underline{v}|} = \frac{\frac{-\sqrt{3}}{2} \underline{i} - \frac{1}{2} \underline{j}}{1} = \frac{-\sqrt{3}}{2} \underline{i} - \frac{1}{2} \underline{j}$ Ans.

Q.7 If A, B and C are respectively the points (2, -4), (4, 0) (1, 6). Use vectors to find coordinates of point D if

(i) ABCD is a parallelogram

Solution:

Let D (x, y) be the required vertex.

Since ABCD is a parallelogram

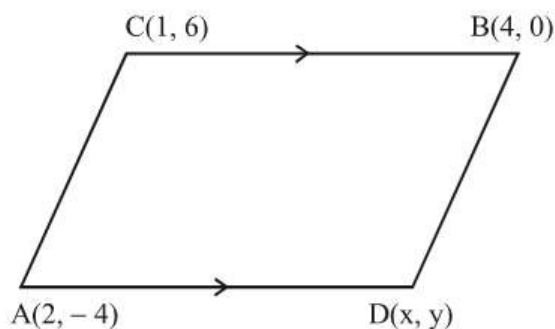
So $\overrightarrow{AB} = \overrightarrow{DC}$

$$(4-2) \underline{i} + (0+4) \underline{j} = (1-x) \underline{i} + (6-y) \underline{j}$$

$$2\underline{i} + 4\underline{j} = (1-x) \underline{i} + (6-y) \underline{j}$$

By comparing

$$2 = 1 - x, \quad 4 = 6 - y$$



$$x = 1 - 2, \quad y = 6 - 4$$

$$x = -1, \quad y = 2$$

Required coordinates of D are $(-1, 2)$

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(ii) **ADBC is a parallelogram.**

Solution:

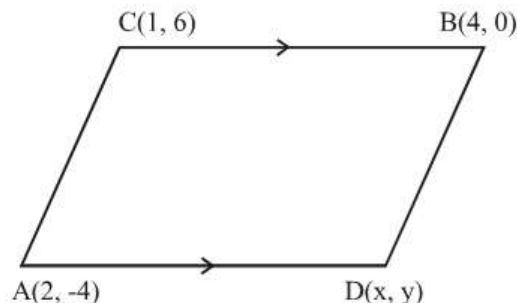
Since ADBC is a parallelogram

$$\begin{aligned}\text{So } \vec{AD} &= \vec{CB} \\ (x-2)\underline{i} + (y+4)\underline{j} &= (4-1)\underline{i} + (0-6)\underline{j} \\ (x-2)\underline{i} + (y+4)\underline{j} &= 3\underline{i} - 6\underline{j}\end{aligned}$$

By comparing

$$\begin{aligned}x-2 &= 3, & y+4 &= -6 \\ x &= 5, & y &= -10\end{aligned}$$

Required coordinates of D are (5, -10)



Q.8 If B, C and D are respectively (4, 1), (-2, 3) & (-8, 0). Use vector method to find the coordinates of the point

(i) **A if ABCD is a parallelogram**

Solution:

Let the coordinates of point A be (x, y)

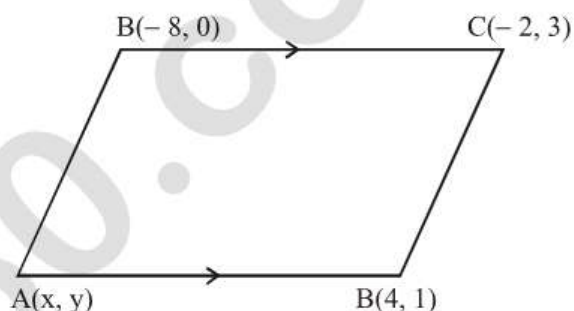
Since ABCD is a parallelogram

$$\begin{aligned}\text{Thus, } \vec{AB} &= \vec{DC} \\ (4-x)\underline{i} + (1-y)\underline{j} &= (-2+8)\underline{i} + (3-0)\underline{j} \\ (4-x)\underline{i} + (1-y)\underline{j} &= 6\underline{i} + 3\underline{j}\end{aligned}$$

By comparing

$$\begin{aligned}4-x &= 6, & 1-y &= 3 \\ 4-6 &= x, & 1-3 &= y \\ -2 &= x, & -2 &= y\end{aligned}$$

Therefore, required point A is (-2, -2)



(ii) **E, if AEBD is a parallelogram**

Solution:

Let the coordinates of E be = (x, y)

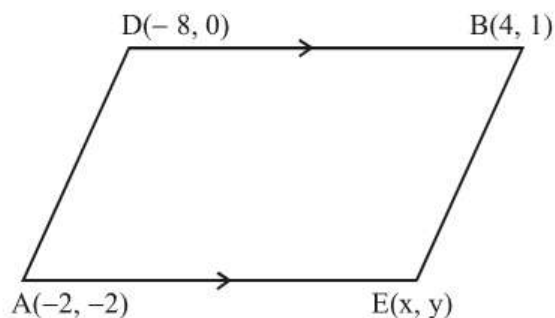
B (4, 1), A (-2, -2), D (-8, 0), E (x, y)

Since AEBD is a parallelogram

$$\begin{aligned}\text{So } \vec{AE} &= \vec{DB} \\ (x+2)\underline{i} + (y+2)\underline{j} &= (4+8)\underline{i} + (1-0)\underline{j} \\ (x+2)\underline{i} + (y+2)\underline{j} &= 12\underline{i} + \underline{j}\end{aligned}$$

By comparing

$$\begin{aligned}x+2 &= 12, & y+2 &= 1 \\ x &= 12-2, & y &= 1-2 \\ x &= 10, & y &= -1\end{aligned}$$



Coordinates of E are (10, -1)

Q.9 If D is origin and $\vec{OP} = \vec{AB}$, find the point, where A and B are (-3, 7) & (1, 0) respectively.

Solution:

Let the coordinates of point P be (x, y)

Therefore

O (0, 0), P (x, y), A (-3, 7), B (1, 0)

Since

$$\vec{OP} = \vec{AB}$$

$$(x - 0)\underline{i} + (y - 0)\underline{j} = (1 + 3)\underline{i} + (0 - 7)\underline{j}$$

$$x\underline{i} + y\underline{j} = 4\underline{i} - 7\underline{j}$$

$$(x, y) = (4, -7) \text{ required point.}$$

Q.10 Use vector to show that ABCD is a parallelogram when the points A, B, C & D are respectively (0, 0), (a, 0), (b, c) & (b - a, c).

(Lahore Board 2009 (supply))

Solution:

Let ABCD be a parallelogram

We have to prove that

$$\vec{AB} = \vec{DC} \quad \& \quad \vec{AD} = \vec{BC}$$

Now

$$\begin{aligned} \vec{AB} &= \vec{OB} - \vec{OA} \\ &= (a - 0)\underline{i} + (0 - 0)\underline{j} = a\underline{i} + 0\underline{j} \quad \dots\dots (i) \end{aligned}$$

$$\begin{aligned} \vec{DC} &= \vec{OC} - \vec{OD} \\ &= (b - b + a)\underline{i} + (c - c)\underline{j} = a\underline{i} + 0\underline{j} \quad \dots\dots (ii) \end{aligned}$$

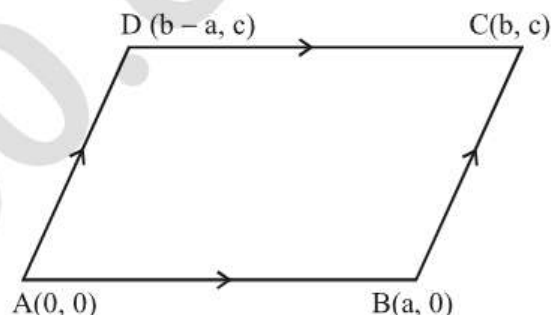
$$\begin{aligned} \vec{AD} &= \vec{OD} - \vec{OA} \\ &= (b - a - 0)\underline{i} + (c - 0)\underline{j} \end{aligned}$$

$$\vec{AD} = (b - a)\underline{i} + c\underline{j} \quad \dots\dots (iii)$$

$$\vec{BC} = \vec{OC} - \vec{OB} = (b - a)\underline{i} + (c - 0)\underline{j}$$

$$\vec{BC} = (b - a)\underline{i} + c\underline{j} \quad \dots\dots (iv)$$

from (i) (ii) (iii) & (iv)



$\vec{AB} = \vec{DC}$ and $\vec{AD} = \vec{BC}$ Shows ABCD is a parallelogram.

Q.11 If $\vec{AB} = \vec{CD}$. Find coordinates of the point A when B, C, D are (1, 2), (-2, 5), D (4, 11) respectively.

Solution:

Let Coordinates of A be (x, y)

A (x, y), B (1, 2), C (-2, 5), D (4, 11)

i.e.; $\vec{AB} = \vec{CD}$

$\vec{OB} - \vec{OA} = \vec{OD} - \vec{OC}$

$(1 - x)\underline{i} + (2 - y)\underline{j} = (4 + 2)\underline{i} + (11 - 5)\underline{j}$

By comparing

$1 - x = 6, \quad 2 - y = 6$

$1 - 6 = x, \quad -y = 6 - 2$

$\Rightarrow x = -5 \quad y = -4$

Hence required point is (-5, -4)

Q.12 Find the position vector of the point of division of the line segments joining the following pair of points.

Formula $\underline{r} = \frac{q\underline{a} + p\underline{b}}{p + q}$

(i) Point C with position vector $2\underline{i} - 3\underline{j}$ and point D with position vector $3\underline{i} + 2\underline{j}$ in ratio 4 : 3. (Lahore Board 2009)

Solution:1

Let the position vector of the required point P be $\underline{r}$ which divides the points C and D in ratio 4:3 By ratio formula

$$\underline{r} = \frac{p\underline{b} + q\underline{a}}{p + q}$$
$$= \frac{3(2\underline{i} - 3\underline{j}) + 4(3\underline{i} + 2\underline{j})}{4 + 3} = \frac{6\underline{i} - 9\underline{j} + 12\underline{i} + 8\underline{j}}{7} = \frac{18\underline{i} - \underline{j}}{7} = \frac{18}{7}\underline{i} - \frac{1}{7}\underline{j}$$

(ii) Point E with position vector $5\underline{i}$ and point F with position vector $4\underline{i} + \underline{j}$ in ratio 2 : 5.

Solution:

Let the position vector of point P be $\underline{r}$ which divides the points E & F in ratio 2:5.



By ratio formula

$$\underline{r} = \frac{P\underline{b} + q\underline{a}}{P + q}$$

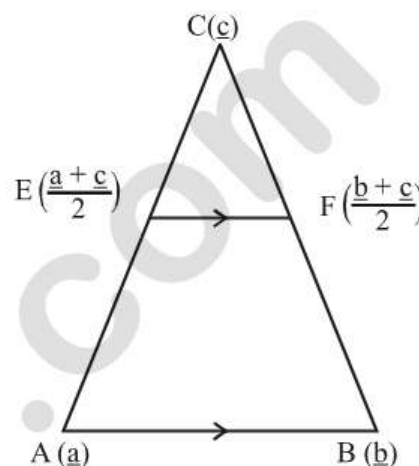


$$\underline{r} = \frac{5(5\underline{i}) + 2(4\underline{i} + \underline{j})}{2 + 5} = \frac{25\underline{i} + 8\underline{i} + 2\underline{j}}{7} = \frac{33\underline{i} + 2\underline{j}}{7} = \frac{33}{7}\underline{i} + \frac{2}{7}\underline{j} \quad \text{Ans.}$$

Q.14 Prove that the line segment joining the mid points of two sides of a triangle is parallel to the third side and half as long. (Lahore Board 2011)

Solution:

Let ABC be any triangle and Let E & F be the mid points of the two sides AC & BC respectively. Let $\underline{a}$, $\underline{b}$, and $\underline{c}$ be position vector of A, B and C. Therefore position vectors of E & F are $\left(\frac{\underline{a} + \underline{c}}{2}\right)$ and $\left(\frac{\underline{b} + \underline{c}}{2}\right)$ respectively.



We have to show that (i) $\vec{AB}$ is parallel to $\vec{EF}$

$$(ii) \quad \frac{1}{2} \vec{AB} = \vec{EF}$$

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$\vec{AB} = \underline{b} - \underline{a} \quad \dots\dots\dots (i)$$

$$\vec{EF} = \vec{OF} - \vec{OE}$$

$$= \frac{\underline{b} + \underline{c}}{2} - \frac{\underline{a} + \underline{c}}{2} = \frac{\underline{b} + \underline{c} - \underline{a} - \underline{c}}{2} = \frac{\underline{b} - \underline{a}}{2}$$

$$\vec{EF} = \frac{1}{2} (\underline{b} - \underline{a}) = \frac{1}{2} \vec{AB} \quad \text{using (i)}$$

$$\vec{EF} = \lambda \vec{AB} \quad \text{where } \lambda = \frac{1}{2}.$$

Hence $\vec{AB}$ and $\vec{EF}$ are parallel & half as long. Hence proved.



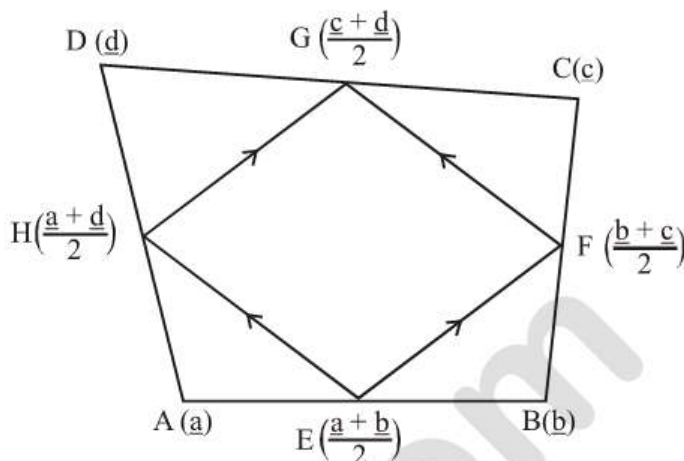
Q.15 Prove that the line segments joining the mid points of the sides of a quadrilateral taken in order form a parallelogram.

(Gujranwala Board 2007, Lahore Board 2009)

Solution:

Let ABCD be any quadrilateral. Let E, F, G, H be mid points of the sides, $\underline{a}$, $\underline{b}$, $\underline{c}$ & $\underline{d}$ are the position vectors of A, B, C and D respectively. The position vectors of E, F, G, & H are $\frac{\underline{a} + \underline{b}}{2}$, $\frac{\underline{b} + \underline{c}}{2}$, $\frac{\underline{c} + \underline{d}}{2}$ & $\frac{\underline{a} + \underline{d}}{2}$ respectively.

We have to prove that EFGH is a parallelogram.



$$\begin{aligned}\vec{EF} &= \vec{OF} - \vec{OE} \\ &= \frac{\underline{b} + \underline{c}}{2} - \frac{\underline{a} + \underline{b}}{2}\end{aligned}$$

$$\vec{EF} = \frac{\underline{b} + \underline{c} - \underline{a} - \underline{b}}{2} = \frac{\underline{c} - \underline{a}}{2} \quad \dots\dots\dots (i)$$

$$\begin{aligned}\vec{HG} &= \vec{OG} - \vec{OH} \\ &= \frac{\underline{c} + \underline{d}}{2} - \frac{\underline{a} + \underline{d}}{2} \\ &= \frac{\underline{c} + \underline{d} - \underline{a} - \underline{d}}{2} = \frac{\underline{c} - \underline{a}}{2} \quad \dots\dots\dots (ii)\end{aligned}$$

$$\begin{aligned}\vec{FG} &= \vec{OG} - \vec{OF} \\ &= \frac{\underline{c} + \underline{d}}{2} - \frac{\underline{b} + \underline{c}}{2} \\ \vec{FG} &= \frac{\underline{c} + \underline{d} - \underline{b} - \underline{c}}{2} = \frac{\underline{d} - \underline{b}}{2} \quad \dots\dots\dots (iii)\end{aligned}$$

$$\begin{aligned}\vec{EH} &= \vec{OH} - \vec{OE} \\ &= \frac{\underline{a} + \underline{d}}{2} - \frac{\underline{a} + \underline{b}}{2} = \frac{\underline{a} + \underline{d} - \underline{a} - \underline{b}}{2} = \frac{\underline{d} - \underline{b}}{2} \quad \dots\dots\dots (iv)\end{aligned}$$

from (i), (ii), (iii) & (iv)

$$\vec{EF} = \vec{HG} \quad \text{and} \quad \vec{EH} = \vec{FG}$$

Shows EFGH is a parallelogram.



EXERCISE 7.2

Q.1 Let $A = (2, 5)$, $B = (-1, 1)$, $C = (2, -6)$ Find (i) $\vec{AB}$

Solution:

$$\begin{aligned}\vec{AB} &= \vec{OB} - \vec{OA} \\ &= (-1-2)\underline{i} + (1-5)\underline{j} = -3\underline{i} - 4\underline{j}\end{aligned}$$

(ii) $2\vec{AB} - \vec{CB}$

Solution:

$$\begin{aligned}2\vec{AB} - \vec{CB} &= 2(\vec{OB} - \vec{OA}) - (\vec{OB} - \vec{OC}) \\ \vec{AB} &= \vec{OB} - \vec{OA} \\ &= -3\underline{i} - 4\underline{j} \\ \vec{CB} &= \vec{OB} - \vec{OC} \\ &= (-1-2)\underline{i} + (1+6)\underline{j} \\ &= -3\underline{i} + 7\underline{j} \\ 2\vec{AB} - \vec{CB} &= 2(-3\underline{i} - 4\underline{j}) - (-3\underline{i} + 7\underline{j}) \\ &= -6\underline{i} - 8\underline{j} + 3\underline{i} - 7\underline{j} = -3\underline{i} - 15\underline{j}\end{aligned}$$

(iii) $2\vec{CB} - 2\vec{CA}$

Solution:

$$\begin{aligned}\vec{CB} &= \vec{OB} - \vec{OC} \\ &= (-1-2)\underline{i} + (1+6)\underline{j} = -3\underline{i} + 7\underline{j} \\ \vec{CA} &= \vec{OA} - \vec{OC} \\ &= (2-2)\underline{i} + (5+6)\underline{j} = 0\underline{i} + 11\underline{j} \\ 2\vec{CB} - 2\vec{CA} &= 2(\vec{CB} - \vec{CA}) \\ &= 2(-3\underline{i} + 7\underline{j} - 0\underline{i} - 11\underline{j}) = 2(-3\underline{i} - 4\underline{j}) = -6\underline{i} - 8\underline{j}\end{aligned}$$



Q.2 Let $\underline{u} = \underline{i} + 2\underline{j} - \underline{k}$, $\underline{v} = 3\underline{i} - 2\underline{j} + 2\underline{k}$
 $\underline{w} = 5\underline{i} - \underline{j} + 3\underline{k}$. Find the indicated vector or number

(i) $\underline{u} + 2\underline{v} + \underline{w}$

Solution:

$$\begin{aligned}\underline{u} + 2\underline{v} + \underline{w} &= (\underline{i} + 2\underline{j} - \underline{k}) + 2(3\underline{i} - 2\underline{j} + 2\underline{k}) + 5\underline{i} - \underline{j} + 3\underline{k} \\ &= \underline{i} + 2\underline{j} - \underline{k} + 6\underline{i} - 4\underline{j} + 4\underline{k} + 5\underline{i} - \underline{j} + 3\underline{k} \\ &= 12\underline{i} - 3\underline{j} + 6\underline{k}\end{aligned}$$

(ii) $\underline{v} - 3\underline{w}$

Solution:

$$\begin{aligned}\underline{v} - 3\underline{w} &= 3\underline{i} - 2\underline{j} + 2\underline{k} - 3(5\underline{i} - \underline{j} + 3\underline{k}) \\ &= 3\underline{i} - 2\underline{j} + 2\underline{k} - 15\underline{i} + 3\underline{j} - 9\underline{k} = -12\underline{i} + \underline{j} - 7\underline{k}\end{aligned}$$

(iii) $|3\underline{v} + \underline{w}|$

Solution:

$$\begin{aligned}3\underline{v} + \underline{w} &= 3(3\underline{i} - 2\underline{j} + 2\underline{k}) + 5\underline{i} - \underline{j} + 3\underline{k} \\ &= 9\underline{i} - 6\underline{j} + 6\underline{k} + 5\underline{i} - \underline{j} + 3\underline{k} \\ &= 14\underline{i} - 7\underline{j} + 9\underline{k}\end{aligned}$$

$$\begin{aligned}|3\underline{v} + \underline{w}| &= \sqrt{(14)^2 + (-7)^2 + (9)^2} \\ &= \sqrt{196 + 49 + 81}\end{aligned}$$

$$|3\underline{v} + \underline{w}| = \sqrt{326} \quad \text{Ans.}$$

Q.3 Find the magnitude of the vector $\underline{v}$ and write the direction cosines of $\underline{v}$.

(i) $\underline{v} = 2\underline{i} + 3\underline{j} + 4\underline{k}$

Solution:

$$\underline{v} = 2\underline{i} + 3\underline{j} + 4\underline{k}$$

$$|\underline{v}| = \sqrt{(2)^2 + (3)^2 + (4)^2} = \sqrt{4 + 9 + 16} = \sqrt{29}$$

direction cosines are

$$\left[\frac{2}{\sqrt{29}}, \frac{3}{\sqrt{29}}, \frac{4}{\sqrt{29}} \right] \quad \text{Ans.}$$



(ii) $\underline{v} = \underline{i} - \underline{j} - \underline{k}$

Solution:

$$\underline{v} = \underline{i} - \underline{j} - \underline{k}$$

$$|\underline{v}| = \sqrt{(1)^2 + (-1)^2 + (-1)^2} = \sqrt{1+1+1} = \sqrt{3}$$

Direction cosines are

$$\left[\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}} \right] \quad \text{Ans.}$$

(iii) $\underline{v} = 4\underline{i} - 5\underline{j}$

Solution:

$$\underline{v} = 4\underline{i} - 5\underline{j}$$

$$|\underline{v}| = \sqrt{(4)^2 + (-5)^2} = \sqrt{16+25} = \sqrt{41}$$

Direction cosines are

$$\left[\frac{4}{\sqrt{41}}, \frac{-5}{\sqrt{41}}, 0 \right] \quad \text{Ans.}$$

Q.4 Find α , so that $|\alpha \underline{i} + (\alpha + 1) \underline{j} + 2\underline{k}| = 3$

(Gujranwala Board 2007)

Solution:

$$|\alpha \underline{i} + (\alpha + 1) \underline{j} + 2\underline{k}| = 3$$

$$\sqrt{\alpha^2 + (\alpha + 1)^2 + (2)^2} = 3$$

Taking square on both sides

$$\alpha^2 + \alpha^2 + 1 + 2\alpha + 4 = 9$$

$$2\alpha^2 + 2\alpha + 5 - 9 = 0$$

$$2\alpha^2 + 2\alpha - 4 = 0$$

$$\alpha^2 + \alpha - 2 = 0 \quad (\text{Dividing throughout by 2})$$

$$\alpha^2 + 2\alpha - \alpha - 2 = 0$$

$$\alpha(\alpha + 2) - 1(\alpha + 2) = 0$$

$$(\alpha + 2)(\alpha - 1) = 0$$

$$\alpha + 2 = 0 \quad \alpha - 1 = 0$$

$$\Rightarrow \alpha = -2, \quad \alpha = 1 \quad \text{Ans}$$

Q.5 Find a unit vector in the direction of $\underline{v} = \underline{i} + 2\underline{j} - \underline{k}$

Solution:

$$\underline{v} = \underline{i} + 2\underline{j} - \underline{k}$$

$$|\underline{v}| = \sqrt{(1)^2 + (2)^2 + (-1)^2} = \sqrt{1+4+1} = \sqrt{6}$$

Required unit vector is

$$\hat{\underline{v}} = \frac{\underline{v}}{|\underline{v}|} = \frac{\underline{i} + 2\underline{j} - \underline{k}}{\sqrt{6}}$$



$$= \frac{1}{\sqrt{6}} \underline{i} + \frac{2}{\sqrt{6}} \underline{j} - \frac{1}{\sqrt{6}} \underline{k} \quad \text{Ans.}$$

Q.6 If $\underline{a} = 3\underline{i} - \underline{j} - 4\underline{k}$, $\underline{b} = -2\underline{i} - 4\underline{j} - 3\underline{k}$ & $\underline{c} = \underline{i} + 2\underline{j} - \underline{k}$. Find a unit vector parallel to $3\underline{a} - 2\underline{b} + 4\underline{c}$ (Gujranwala Board 2004)

Solution:

$$3\underline{a} = 3(3\underline{i} - \underline{j} - 4\underline{k}) = 9\underline{i} - 3\underline{j} - 12\underline{k}$$

$$2\underline{b} = 2(-2\underline{i} - 4\underline{j} - 3\underline{k}) = -4\underline{i} - 8\underline{j} - 6\underline{k}$$

$$4\underline{c} = 4(\underline{i} + 2\underline{j} - \underline{k}) = 4\underline{i} + 8\underline{j} - 4\underline{k}$$

$$\text{Let } \underline{v} = 3\underline{a} - 2\underline{b} + 4\underline{c} = 9\underline{i} - 3\underline{j} - 12\underline{k} - (-4\underline{i} - 8\underline{j} - 6\underline{k}) + 4\underline{i} + 8\underline{j} - 4\underline{k}$$

$$= 9\underline{i} - 3\underline{j} - 12\underline{k} + 4\underline{i} + 8\underline{j} + 6\underline{k} + 4\underline{i} + 8\underline{j} - 4\underline{k}$$

$$\underline{v} = 17\underline{i} + 13\underline{j} - 10\underline{k}$$

$$\text{Now } |\underline{v}| = \sqrt{(17)^2 + (13)^2 + (-10)^2} = \sqrt{289 + 169 + 100} = \sqrt{558}$$

$$\hat{\underline{v}} = \frac{\underline{v}}{|\underline{v}|} = \frac{17\underline{i} + 13\underline{j} - 10\underline{k}}{\sqrt{558}} = \frac{17}{\sqrt{558}} \underline{i} + \frac{13}{\sqrt{558}} \underline{j} - \frac{10}{\sqrt{558}} \underline{k} \quad \text{Ans.}$$

Q.7 Find a vector whose

(i) magnitude is 4 and is parallel to $2\underline{i} - 3\underline{j} + 6\underline{k}$

Solution:

$$\text{Let } \underline{v} = 2\underline{i} - 3\underline{j} + 6\underline{k}$$

$$|\underline{v}| = \sqrt{(2)^2 + (-3)^2 + (6)^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$$

Let $\underline{u}$ be a vector parallel to $\underline{v}$, then

$$\underline{u} = \frac{\underline{v}}{|\underline{v}|} = \frac{2\underline{i} - 3\underline{j} + 6\underline{k}}{7} \quad (\text{It is a vector whose magnitude is 1 and parallel to } \underline{v})$$

Required vector

$$4\underline{u} = 4 \left(\frac{2\underline{i} - 3\underline{j} + 6\underline{k}}{7} \right) = \frac{8}{7} \underline{i} - \frac{12}{7} \underline{j} + \frac{24}{7} \underline{k} \quad \text{Ans.}$$

(ii) magnitude is 2 and is parallel to $-\underline{i} + \underline{j} + \underline{k}$ (Lahore Board 2006)

Solution:

$$\text{Let } \underline{v} = -\underline{i} + \underline{j} + \underline{k}$$

$$|\underline{v}| = \sqrt{(-1)^2 + (1)^2 + (1)^2} = \sqrt{3}$$

Let $\underline{u}$ is vector parallel to $\underline{v}$

$$\underline{u} = \frac{\underline{v}}{|\underline{v}|} = \frac{-\underline{i} + \underline{j} + \underline{k}}{\sqrt{3}}$$



Required vector

$$2\bar{u} = \frac{2(-\bar{i} + \bar{j} + \bar{k})}{\sqrt{3}} = \frac{-2}{\sqrt{3}}\bar{i} + \frac{2}{\sqrt{3}}\bar{j} + \frac{2}{\sqrt{3}}\bar{k} \quad \text{Ans.}$$

Q.8 If $\bar{u} = 2\bar{i} + 3\bar{j} + 4\bar{k}$, $\bar{v} = -\bar{i} + 3\bar{j} - \bar{k}$, $\bar{w} = \bar{i} + 6\bar{j} + Z\bar{k}$ represents the sides of a triangle. Find the value of Z.

Solution:

It $\bar{u}$, $\bar{v}$ & $\bar{w}$ represents the sides of a triangle, then by vector addition $\bar{u} + \bar{v} = \bar{w}$

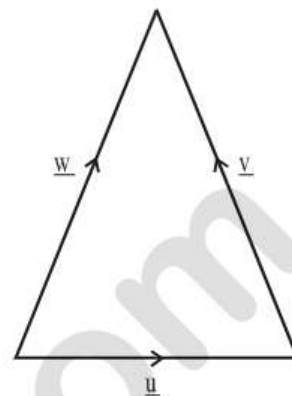
$$2\bar{i} + 3\bar{j} + 4\bar{k} + (-\bar{i} + 3\bar{j} - \bar{k}) = \bar{i} + 6\bar{j} + Z\bar{k}$$

$$2\bar{i} + 3\bar{j} + 4\bar{k} - \bar{i} + 3\bar{j} - \bar{k} = \bar{i} + 6\bar{j} + Z\bar{k}$$

$$\bar{i} + 6\bar{j} + 3\bar{k} = \bar{i} + 6\bar{j} + Z\bar{k}$$

By comparing

$$Z = 3 \quad \text{Ans.}$$



Q.9 The position vectors of the points A, B, C and D are $2\bar{i} - \bar{j} + \bar{k}$, $3\bar{i} + \bar{j}$, $2\bar{i} + 4\bar{j} - 2\bar{k}$ and $-\bar{i} - 2\bar{j} + \bar{k}$ respectively. Show that $\vec{AB}$ is parallel to $\vec{CD}$.

Solution:

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$= (3 - 2)\bar{i} + (1 + 1)\bar{j} + (0 - 1)\bar{k}$$

$$\vec{AB} = \bar{i} + 2\bar{j} - \bar{k}$$

$$\vec{CD} = \text{Position vector of D} - \text{Position vector of C}$$

$$= (-1 - 2)\bar{i} + (-2 - 4)\bar{j} + (1 + 2)\bar{k}$$

$$= -3\bar{i} - 6\bar{j} + 3\bar{k}$$

$$\vec{CD} = -3(\bar{i} + 2\bar{j} - \bar{k})$$

$$\vec{CD} = -3\vec{AB}$$

Hence $\vec{AB}$ is parallel to $\vec{CD}$.



Q.10 Two vectors $\underline{u}$ & $\underline{w}$ in space are parallel, if there is a scalar c such that $\underline{v} = c\underline{w}$. The vectors point in the same direction if $c > 0$ and the vector point in the opposite direction if $c < 0$

(a) Find two vectors of length 2 parallel to vector $\underline{v} = 2\underline{i} - 4\underline{j} + 4\underline{k}$

Solution:

$$\begin{aligned}\underline{v} &= 2\underline{i} - 4\underline{j} + 4\underline{k} \\ |\underline{v}| &= \sqrt{(2)^2 + (-4)^2 + (4)^2} = \sqrt{4 + 16 + 16} = \sqrt{36} = 6 \\ \Rightarrow \hat{\underline{v}} &= \frac{\underline{v}}{|\underline{v}|} = \frac{2\underline{i} - 4\underline{j} + 4\underline{k}}{6} = \frac{2(\underline{i} - 2\underline{j} + 2\underline{k})}{6} = \frac{\underline{i} - 2\underline{j} + 2\underline{k}}{3}\end{aligned}$$

$\therefore$ The two vectors whose length is 2 and parallel to $\hat{\underline{v}}$ are $2\hat{\underline{v}}$ & $-2\hat{\underline{v}}$

i.e; $2\hat{\underline{v}} = \frac{2}{3} (\underline{i} - 2\underline{j} + 2\underline{k}) = \frac{2}{3}\underline{i} - \frac{4}{3}\underline{j} + \frac{4}{3}\underline{k}$ Ans.

$-2\hat{\underline{v}} = \frac{-2}{3} (\underline{i} - 2\underline{j} + 2\underline{k}) = \frac{-2}{3}\underline{i} + \frac{4}{3}\underline{j} - \frac{4}{3}\underline{k}$ Ans.

(b) Find the constant a so that the vectors $\underline{v} = \underline{i} - 3\underline{j} + 4\underline{k}$ and $\underline{w} = a\underline{i} + 9\underline{j} - 12\underline{k}$ are parallel. (Gujranwala Board 2004)

Solution:

Since $\underline{v}$ & $\underline{w}$ are parallel so

$$\underline{w} = c\underline{v}$$

$$a\underline{i} + 9\underline{j} - 12\underline{k} = c(\underline{i} - 3\underline{j} + 4\underline{k})$$

$$a\underline{i} + 9\underline{j} - 12\underline{k} = c\underline{i} - 3c\underline{j} + 4c\underline{k}$$

By comparing

$$a = c, \quad 9 = -3c, \quad -12 = 4c$$

$$\Rightarrow \frac{9}{-3} = c \quad \Rightarrow c = -3$$

$$\boxed{a = -3} \quad \text{Ans.}$$

(c) Find a vector of length 5 in the direction opposite that of $\underline{v} = \underline{i} - 2\underline{j} + 3\underline{k}$. (Lahore Board 2004)

Solution:

$$\begin{aligned}\underline{v} &= \underline{i} - 2\underline{j} + 3\underline{k} \\ |\underline{v}| &= \sqrt{(1)^2 + (-2)^2 + (3)^2} = \sqrt{1 + 4 + 9} = \sqrt{14} \\ \hat{\underline{v}} &= \frac{\underline{v}}{|\underline{v}|} = \frac{\underline{i} - 2\underline{j} + 3\underline{k}}{\sqrt{14}}\end{aligned}$$

$\therefore$ The vector of length 5 in opposite direction of $\underline{v}$ is



$$\begin{aligned} -5\hat{v} &= \frac{-5}{\sqrt{14}} (\underline{i} - 2\underline{j} + 3\underline{k}) \\ \frac{-5}{\sqrt{14}} \underline{i} + \frac{10}{\sqrt{14}} \underline{j} - \frac{15}{\sqrt{14}} \underline{k} &\quad \text{Ans.} \end{aligned}$$

(d) Find a and b so that the vectors $3\underline{i} - \underline{j} + 4\underline{k}$ and $a\underline{i} + b\underline{j} - 2\underline{k}$ are parallel.

Solution:

Since $\underline{v}$ & $\underline{w}$ are parallel so

$$\underline{w} = c\underline{v}$$

$$a\underline{i} + b\underline{j} - 2\underline{k} = c(3\underline{i} - \underline{j} + 4\underline{k})$$

$$a\underline{i} + b\underline{j} - 2\underline{k} = 3c\underline{i} - c\underline{j} + 4c\underline{k}$$

By comparing

$$\begin{aligned} a &= 3c, & b &= -c, & -2 &= 4c \\ \frac{-2}{4} &= c \end{aligned}$$

$$b = -c$$

$$\boxed{\frac{-1}{2} = c}$$

$$\Rightarrow \boxed{b = \frac{1}{2}} \quad a = 3c \Rightarrow a = 3\left(\frac{-1}{2}\right) \Rightarrow \boxed{a = \frac{-3}{2}}$$

Q.11 Find the direction cosines for the given vectors.

(i) $\underline{v} = 3\underline{i} - \underline{j} + 2\underline{k}$ (Lahore Board 2007)

Solution:

$$\begin{aligned} \underline{v} &= 3\underline{i} - \underline{j} + 2\underline{k} \\ \frac{\underline{v}}{|\underline{v}|} &= \frac{3\underline{i} - \underline{j} + 2\underline{k}}{\sqrt{(3)^2 + (-1)^2 + (2)^2}} = \frac{3\underline{i} - \underline{j} + 2\underline{k}}{\sqrt{9 + 1 + 4}} = \frac{3\underline{i} - \underline{j} + 2\underline{k}}{\sqrt{14}} \end{aligned}$$

Direction cosines are

$$= \left[\frac{3}{\sqrt{14}}, \frac{-1}{\sqrt{14}}, \frac{2}{\sqrt{14}} \right]$$

(ii) $\underline{v} = 6\underline{i} - 2\underline{j} + \underline{k}$ (Lahore Board 2006)

Solution:

$$\begin{aligned} \underline{v} &= 6\underline{i} - 2\underline{j} + \underline{k} \\ \frac{\underline{v}}{|\underline{v}|} &= \frac{6\underline{i} - 2\underline{j} + \underline{k}}{\sqrt{(6)^2 + (-2)^2 + (1)^2}} = \frac{6\underline{i} - 2\underline{j} + \underline{k}}{\sqrt{36 + 4 + 1}} = \frac{6\underline{i} - 2\underline{j} + \underline{k}}{\sqrt{41}} \end{aligned}$$

$$\text{Direction cosines are } = \left[\frac{6}{\sqrt{41}}, \frac{-2}{\sqrt{41}}, \frac{1}{\sqrt{41}} \right] \quad \text{Ans.}$$



(iii) $\vec{PQ}$, where P (2, 1, 5) & Q = (1, 3, 1)

Solution:

$$\begin{aligned}\vec{PQ} &= \vec{OQ} - \vec{OP} \\ &= (1 - 2)\underline{i} + (3 - 1)\underline{j} + (1 - 5)\underline{k} = -\underline{i} + 2\underline{j} - 4\underline{k}\end{aligned}$$

$$|\vec{PQ}| = \sqrt{(-1)^2 + (2)^2 + (-4)^2} = \sqrt{1 + 4 + 16} = \sqrt{21}$$

$$\text{Direction cosines are} = \left[\frac{-1}{\sqrt{21}}, \frac{2}{\sqrt{21}}, \frac{-4}{\sqrt{21}} \right] \quad \text{Ans.}$$

Q.12 Which of the following triples can be the direction angles of a single vector.

(i) $45^\circ, 45^\circ, 60^\circ$

Solution:

If α, β, γ are direction angles of a vector, then it must satisfy $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

L.H.S.

$$\begin{aligned}\cos^2\alpha + \cos^2\beta + \cos^2\gamma &= (\cos 45^\circ)^2 + (\cos 45^\circ)^2 + (\cos 60^\circ)^2 \\ &= \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2} + \frac{1}{2} + \frac{1}{4} \\ &= \frac{2 + 2 + 1}{4} = \frac{5}{4} \neq 1\end{aligned}$$

So given triples are not direction angles.

(ii) $30^\circ, 45^\circ, 60^\circ$

Solution:

$$\begin{aligned}\alpha &= 30^\circ, \beta = 45^\circ, \gamma = 60^\circ \\ \cos^2\alpha + \cos^2\beta + \cos^2\gamma &= (\cos 30^\circ)^2 + (\cos 45^\circ)^2 + (\cos 60^\circ)^2 \\ &= \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{3}{4} + \frac{1}{2} + \frac{1}{4} = \frac{3 + 2 + 1}{4} = \frac{6}{4} \neq 1\end{aligned}$$

Hence given triples can not be direction angles.

(iii) $45^\circ, 60^\circ, 60^\circ$

Solution:

$$\begin{aligned}\alpha &= 45^\circ, \beta = 60^\circ, \gamma = 60^\circ \\ \cos^2\alpha + \cos^2\beta + \cos^2\gamma &= (\cos 45^\circ)^2 + (\cos 60^\circ)^2 + (\cos 60^\circ)^2 \\ &= \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} \\ &= \frac{2 + 1 + 1}{4} = \frac{4}{4} = 1\end{aligned}$$



As $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Therefore, given triples can be direction angles of a vector.

The Scalar Product of Two vectors

Definition:

Let two non zero vectors $\underline{u}$ & $\underline{v}$ in the plane or in space, have same initial point.

The dot product of $\underline{u}$ and $\underline{v}$, written as $\underline{u} \cdot \underline{v}$, is defined by

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos\theta \text{ where } \theta \text{ is angle between } \underline{u} \text{ \& } \underline{v} \text{ and } 0 \leq \theta \leq \pi.$$

Orthogonal / Perpendicular vectors:

The two vectors $\underline{u}$ & $\underline{v}$ are orthogonal / perpendicular if and only if $\underline{u} \cdot \underline{v} = 0$

Remember:

- (i) Dot product, inner product, scalar product are same.
- (ii) $\underline{i} \cdot \underline{i} = \underline{j} \cdot \underline{j} = \underline{k} \cdot \underline{k} = 1$
- (iii) $\underline{i} \cdot \underline{j} = \underline{j} \cdot \underline{k} = \underline{k} \cdot \underline{i} = 0$
- (iv) Scalar product is commutative i.e., $\underline{u} \cdot \underline{v} = \underline{v} \cdot \underline{u}$



EXERCISE 7.3

Q.1 Find the Cosine of the angle θ between $\underline{u}$ and $\underline{v}$.

(i) $\underline{u} = 3\underline{i} + \underline{j} - \underline{k}$ $\underline{v} = 2\underline{i} - \underline{j} + \underline{k}$

Formula

$$\cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

Solution:

$$\underline{u} = 3\underline{i} + \underline{j} - \underline{k} \quad , \quad \underline{v} = 2\underline{i} - \underline{j} + \underline{k}$$

$$\underline{u} \cdot \underline{v} = (3\underline{i} + \underline{j} - \underline{k}) \cdot (2\underline{i} - \underline{j} + \underline{k})$$

$$\underline{u} \cdot \underline{v} = 6 - 1 - 1 = 4$$

$$|\underline{u}| = \sqrt{(3)^2 + (1)^2 + (-1)^2} = \sqrt{9 + 1 + 1} = \sqrt{11}$$

$$|\underline{v}| = \sqrt{(2)^2 + (-1)^2 + (1)^2} = \sqrt{4 + 1 + 1} = \sqrt{6}$$

$$\cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|} = \frac{4}{\sqrt{11} \sqrt{6}} = \frac{4}{\sqrt{66}}$$

$$\cos \theta = \frac{4}{\sqrt{66}} \quad \text{Ans.}$$



(ii) $\underline{u} = \underline{i} - 3\underline{j} + 4\underline{k}$, $\underline{v} = 4\underline{i} - \underline{j} + 3\underline{k}$

Solution:

$$\begin{aligned}\underline{u} \cdot \underline{v} &= (\underline{i} - 3\underline{j} + 4\underline{k}) \cdot (4\underline{i} - \underline{j} + 3\underline{k}) \\ &= 4 + 3 + 12\end{aligned}$$

$$\underline{u} \cdot \underline{v} = 19$$

$$|\underline{u}| = \sqrt{(1)^2 + (-3)^2 + (4)^2} = \sqrt{1 + 9 + 16} = \sqrt{26}$$

$$|\underline{v}| = \sqrt{(4)^2 + (-1)^2 + (3)^2} = \sqrt{16 + 1 + 9} = \sqrt{26}$$

$$\cos\theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|} = \frac{19}{\sqrt{26} \sqrt{26}}$$

$$\cos\theta = \frac{19}{26} \quad \text{Ans.}$$

(iii) $\underline{u} = [-3, 5]$, $\underline{v} = [6, -2]$

Solution:

$$\underline{u} = -3\underline{i} + 5\underline{j}, \quad \underline{v} = 6\underline{i} - 2\underline{j}$$

$$\begin{aligned}\underline{u} \cdot \underline{v} &= (-3\underline{i} + 5\underline{j}) \cdot (6\underline{i} - 2\underline{j}) \\ &= -18 - 10\end{aligned}$$

$$\underline{u} \cdot \underline{v} = -28$$

$$|\underline{u}| = \sqrt{(-3)^2 + (5)^2} = \sqrt{9 + 25} = \sqrt{34}$$

$$|\underline{v}| = \sqrt{(6)^2 + (-2)^2} = \sqrt{36 + 4} = \sqrt{40}$$

$$\cos\theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$\cos\theta = \frac{-28}{\sqrt{34} \sqrt{40}} = \frac{-28}{\sqrt{34} \sqrt{10}} = \frac{-28}{\sqrt{2 \times 17} \sqrt{2 \times 5}} = \frac{-14}{2\sqrt{85}}$$

$$\cos\theta = \frac{-7}{\sqrt{85}} \quad \text{Ans.}$$

(iv) $\underline{u} = [2, -3, 1]$, $\underline{v} = [2, 4, 1]$

Solution:

$$\underline{u} = 2\underline{i} - 3\underline{j} + \underline{k}, \quad \underline{v} = 2\underline{i} + 4\underline{j} + \underline{k}$$

$$\begin{aligned}\underline{u} \cdot \underline{v} &= (2\underline{i} - 3\underline{j} + \underline{k}) \cdot (2\underline{i} + 4\underline{j} + \underline{k}) \\ &= 4 - 12 + 1\end{aligned}$$

$$\underline{u} \cdot \underline{v} = -7$$

$$|\underline{u}| = \sqrt{(2)^2 + (-3)^2 + (1)^2} = \sqrt{4 + 9 + 1} = \sqrt{14}$$

$$|\underline{v}| = \sqrt{(2)^2 + (4)^2 + (1)^2} = \sqrt{4 + 16 + 1} = \sqrt{21}$$



$$\begin{aligned}\cos\theta &= \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|} \\ \cos\theta &= \frac{-7}{\sqrt{14}\sqrt{21}} = \frac{-7}{\sqrt{2 \times 7 \times 3 \times 7}} = \frac{-7}{7\sqrt{6}} = \frac{-1}{\sqrt{6}} \\ \cos\theta &= \frac{-1}{\sqrt{6}} \quad \text{Ans.}\end{aligned}$$

Q.2 Calculate the projection of $\underline{a}$ along $\underline{b}$ and projection of $\underline{b}$ along $\underline{a}$ when

(i) $\underline{a} = \underline{i} - \underline{k}$, $\underline{b} = \underline{j} + \underline{k}$

Solution:

Formula Projection of $\underline{a}$ along $\underline{b} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|}$

$$\text{Project of } \underline{b} \text{ along } \underline{a} = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}|}$$

$$\underline{a} = \underline{i} + 0\underline{j} - \underline{k}, \quad \underline{b} = 0\underline{i} + \underline{j} + \underline{k}$$

$$\begin{aligned}\underline{a} \cdot \underline{b} &= (\underline{i} + 0\underline{j} - \underline{k}) \cdot (0\underline{i} + \underline{j} + \underline{k}) \\ &= 0 + 0 - 1\end{aligned}$$

$$\underline{a} \cdot \underline{b} = -1$$

$$|\underline{a}| = \sqrt{(1)^2 + 0 + (-1)^2} = \sqrt{1+1} = \sqrt{2}$$

$$|\underline{b}| = \sqrt{(1)^2 + (1)^2} = \sqrt{1+1} = \sqrt{2}$$

$$\text{Projection of } \underline{a} \text{ along } \underline{b} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|} = \frac{-1}{\sqrt{2}} \quad \text{Ans.}$$

$$\text{And Projection of } \underline{b} \text{ along } \underline{a} = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}|} = \frac{-1}{\sqrt{2}} \quad \text{Ans.}$$

(ii) $\underline{a} = 3\underline{i} + \underline{j} - \underline{k}$, $\underline{b} = 2\underline{i} - \underline{j} + \underline{k}$ (Gujranwala Board 2004, 2007)

Solution:

$$\underline{a} \cdot \underline{b} = (3\underline{i} + \underline{j} - \underline{k}) \cdot (2\underline{i} - \underline{j} + \underline{k})$$

$$\underline{a} \cdot \underline{b} = -6 - 1 - 1 = -8$$

$$|\underline{a}| = \sqrt{(3)^2 + (1)^2 + (-1)^2} = \sqrt{9+1+1} = \sqrt{11}$$

$$|\underline{b}| = \sqrt{(2)^2 + (-1)^2 + (1)^2} = \sqrt{4+1+1} = \sqrt{6}$$

$$\text{Projection of } \underline{a} \text{ along } \underline{b} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|} = \frac{-8}{\sqrt{6}} \quad \text{Ans.}$$

$$\text{Projection of } \underline{b} \text{ along } \underline{a} = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}|} = \frac{-8}{\sqrt{11}} \quad \text{Ans.}$$



Q.3 Find a real number α so that the vectors $\underline{u}$ & $\underline{v}$ are perpendicular.

(i) $\underline{u} = 2\alpha \underline{i} + \underline{j} - \underline{k}$ $\underline{v} = \underline{i} + \alpha \underline{j} + 4\underline{k}$ (Lahore Board 2010,11)

Solution:

Since $\underline{u}$ & $\underline{v}$ are perpendicular so

$$\underline{u} \cdot \underline{v} = 0$$

$$(2\alpha \underline{i} + \underline{j} - \underline{k}) \cdot (\underline{i} + \alpha \underline{j} + 4\underline{k}) = 0$$

$$2\alpha + \alpha - 4 = 0$$

$$3\alpha - 4 = 0$$

$$\alpha = \frac{4}{3} \quad \text{Ans.}$$

(i) $\underline{u} = \alpha \underline{i} + 2\alpha \underline{j} - \underline{k}$ $\underline{v} = \underline{i} + \alpha \underline{j} + 3\underline{k}$ (Lahore Board 2006)

Solution:

$$\underline{u} = \alpha \underline{i} + 2\alpha \underline{j} - \underline{k}, \quad \underline{v} = \underline{i} + \alpha \underline{j} + 3\underline{k}$$

Since $\underline{u}$ & $\underline{v}$ are perpendicular so

$$\underline{u} \cdot \underline{v} = 0$$

$$(\alpha \underline{i} + 2\alpha \underline{j} - \underline{k}) \cdot (\underline{i} + \alpha \underline{j} + 3\underline{k}) = 0$$

$$\alpha + 2\alpha^2 - 3 = 0$$

$$2\alpha^2 + \alpha - 3 = 0$$

$$2\alpha^2 + 3\alpha - 2\alpha - 3 = 0$$

$$\alpha(2\alpha + 3) - 1(2\alpha + 3) = 0$$

$$(\alpha - 1)(2\alpha + 3) = 0$$

$$\alpha - 1 = 0, \quad 2\alpha + 3 = 0$$

$$\alpha = 1, \quad \alpha = -\frac{3}{2} \quad \text{Ans.}$$

Q.4 Find the number Z so that the triangle with vertices A (1, -1, 0), B (-2, 2, 1) and C(0, 2, Z) is a right triangle with right angle at C.

Solution:

Given A (1, -1, 0), B (-2, 2, 1), C (0, 2, Z)

$$\begin{aligned} \vec{AC} &= \vec{OC} - \vec{OA} \\ &= (0 - 1)\underline{i} + (2 + 1)\underline{j} + (Z - 0)\underline{k} \end{aligned}$$

$$\vec{AC} = -\underline{i} + 3\underline{j} + Z\underline{k}$$

$$\begin{aligned} \vec{BC} &= \vec{OC} - \vec{OB} \\ &= (0 + 2)\underline{i} + (2 - 2)\underline{j} + (Z - 1)\underline{k} \end{aligned}$$



$$\vec{BC} = 2\vec{i} + 0\vec{j} + (Z - 1)\vec{k}$$

Since $\vec{AC}$ & $\vec{BC}$ are perpendicular
So,

$$\vec{AC} \cdot \vec{BC} = 0$$

$$(-\vec{i} + 3\vec{j} + Z\vec{k}) \cdot (2\vec{i} + 0\vec{j} + (Z - 1)\vec{k}) = 0$$

$$-2 + 0 + Z(Z - 1) = 0$$

$$-2 + Z^2 - Z = 0$$

$$Z^2 - Z - 2 = 0$$

$$Z^2 - 2Z + Z - 2 = 0$$

$$Z(Z - 2) + 1(Z - 2) = 0$$

$$(Z - 2)(Z + 1) = 0$$

$$\Rightarrow Z - 2 = 0 \quad Z + 1 = 0$$

$$Z = 2, \quad Z = -1$$

Q.5 If $\underline{V}$ is a vector for which

$$\underline{v} \cdot \underline{i} = 0 \quad \underline{v} \cdot \underline{j} = 0, \quad \underline{v} \cdot \underline{k} = 0, \text{ find } \underline{v}$$

(Lahore Board 2009)

Solution:

$$\text{Let } \underline{v} = x\underline{i} + y\underline{j} + z\underline{k} \quad \dots\dots\dots (1)$$

Now

$$\underline{v} \cdot \underline{i} = 0$$

$$(x\underline{i} + y\underline{j} + z\underline{k}) \cdot (\underline{i}) = 0$$

$$x = 0$$

Next

$$\underline{v} \cdot \underline{j} = 0$$

$$(x\underline{i} + y\underline{j} + z\underline{k}) \cdot (0\underline{i} + \underline{j} + 0\underline{k}) = 0$$

$$0 + y + 0 = 0 \quad \Rightarrow y = 0$$

$$\underline{v} \cdot \underline{k} = 0$$

$$(x\underline{i} + y\underline{j} + z\underline{k}) \cdot (0\underline{i} + \underline{j} + 0\underline{k}) = 0$$

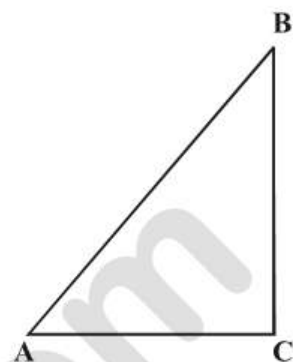
$$0 + 0 + z = 0$$

$$\Rightarrow z = 0$$

Substitute all values in (1)

$$\underline{v} = 0\underline{i} + 0\underline{j} + 0\underline{k}$$

$$\Rightarrow \underline{v} = 0 \quad (\text{Null vector}) \quad \text{Ans.}$$



Q.6(i) Show that the vectors $3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$ & $2\mathbf{i} + \mathbf{j} - 4\mathbf{k}$ form a right angle triangle.

Solution:

$$\text{Let } \underline{u} = 3\underline{i} - 2\underline{j} + \underline{k}, \quad \underline{v} = \underline{i} - 3\underline{j} + 5\underline{k}, \quad \underline{w} = 2\underline{i} + \underline{j} - 4\underline{k}$$

$$\begin{aligned} \underline{v} + \underline{w} &= \underline{i} - 3\underline{j} + 5\underline{k} + 2\underline{i} + \underline{j} - 4\underline{k} \\ &= 3\underline{i} - 2\underline{j} + \underline{k} \end{aligned}$$

$$\underline{v} + \underline{w} = \underline{u}$$

Hence $\underline{u}, \underline{v}, \underline{w}$ form a triangle

$$\begin{aligned} \underline{u} \cdot \underline{w} &= (3\underline{i} - 2\underline{j} + \underline{k}) \cdot (2\underline{i} + \underline{j} - 4\underline{k}) \\ &= 6 - 2 - 4 = 0 \end{aligned}$$

$\Rightarrow$ $\underline{u}$ and $\underline{w}$ are perpendicular to each other.

Therefore, given triangle is right angled triangle.

(ii) Show that the set of points P(1, 3, 2), Q (4, 1, 4). R (6, 5, 5) form a right triangle.

Solution:

$$\begin{aligned} \overrightarrow{PQ} &= \overrightarrow{OQ} - \overrightarrow{OP} \\ &= (4-1)\underline{i} + (1-3)\underline{j} + (4-2)\underline{k} = 3\underline{i} - 2\underline{j} + 2\underline{k} \end{aligned}$$

$$\overrightarrow{QR} = (6-4)\underline{i} + (5-1)\underline{j} + (5-4)\underline{k} = 2\underline{i} + 4\underline{j} + \underline{k}$$

$$\overrightarrow{PR} = (6-1)\underline{i} + (5-3)\underline{j} + (5-2)\underline{k} = 5\underline{i} + 2\underline{j} + 3\underline{k}$$

$$\overrightarrow{QR} = \overrightarrow{PR}$$

$$\begin{aligned} \text{Now } \overrightarrow{PQ} + \overrightarrow{QR} &= 3\underline{i} - 2\underline{j} + 2\underline{k} + 2\underline{i} + 4\underline{j} + \underline{k} \\ &= 5\underline{i} + 2\underline{j} + 3\underline{k} \end{aligned}$$

$$\overrightarrow{PQ} + \overrightarrow{QR} = \overrightarrow{PR}$$

Thus,

P, Q, R form a triangle

$$\begin{aligned} \overrightarrow{PQ} \cdot \overrightarrow{QR} &= (3\underline{i} - 2\underline{j} + 2\underline{k}) \cdot (2\underline{i} + 4\underline{j} + \underline{k}) \\ &= 6 - 8 + 2 = 0 \end{aligned}$$

Therefore $\overrightarrow{PQ}$ & $\overrightarrow{QR}$ are perpendicular to each other

Thus, given triangle is right-angled triangle.



Q.7 Show that mid point of hypotenuse a right triangle is equidistant from its vertices.

Solution:

Let AOB be any triangle with vertex O is at origin.

Therefore, coordinates of O, A, and B will be O (0, 0), A (a, 0) B (0, b).

Coordinates of mid point M are =

$$\left(\frac{a+0}{2}, \frac{0+b}{2}\right) = \left(\frac{a}{2}, \frac{b}{2}\right)$$

We have to prove that mid point of hypotenuse is equidistant from its vertices i.e.

$$|\vec{OM}| = |\vec{AM}| = |\vec{BM}|$$

$$\vec{OM} = \left(\frac{a}{2} - 0\right)\underline{i} + \left(\frac{b}{2} - 0\right)\underline{j} = \frac{a}{2}\underline{i} + \frac{b}{2}\underline{j}$$

$$|\vec{OM}| = \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2} = \sqrt{\frac{a^2}{4} + \frac{b^2}{4}} = \sqrt{\frac{a^2 + b^2}{4}} = \frac{\sqrt{a^2 + b^2}}{2} \quad \text{--- (i)}$$

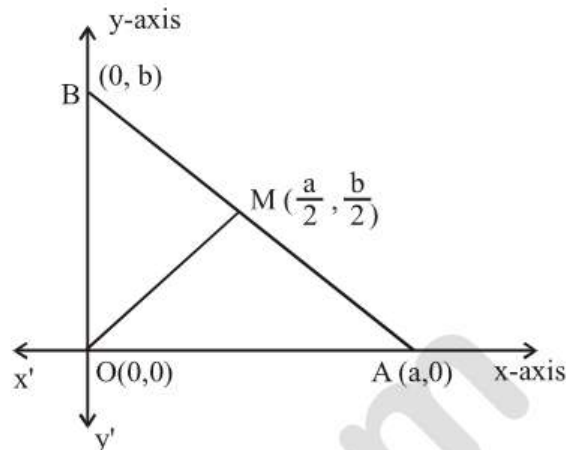
$$\vec{AM} = \left(\frac{a}{2} - a\right)\underline{i} + \left(\frac{b}{2} - 0\right)\underline{j} = \frac{-a}{2}\underline{i} + \frac{b}{2}\underline{j}$$

$$\begin{aligned} |\vec{AM}| &= \sqrt{\left(\frac{-a}{2}\right)^2 + \left(\frac{b}{2}\right)^2} = \sqrt{\frac{a^2}{4} + \frac{b^2}{4}} \\ &= \sqrt{\frac{a^2 + b^2}{4}} = \frac{\sqrt{a^2 + b^2}}{2} \quad \text{--- (ii)} \end{aligned}$$

$$\vec{BM} = \left(\frac{a}{2} - 0\right)\underline{i} + \left(\frac{b}{2} - b\right)\underline{j} = \frac{a}{2}\underline{i} - \frac{b}{2}\underline{j}$$

$$|\vec{BM}| = \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{-b}{2}\right)^2} = \sqrt{\frac{a^2}{4} + \frac{b^2}{4}} = \sqrt{\frac{a^2 + b^2}{4}} = \frac{\sqrt{a^2 + b^2}}{2} \quad \text{--- (iii)}$$

From (i) (ii) & (iii) M is equidistant from its vertices.



Q.8 Prove that perpendicular bisectors of the sides of a triangle are concurrent.

Solution:

Let $\vec{OD}$ & $\vec{OE}$ be the perpendicular bisectors of the sides $\vec{AB}$ and $\vec{BC}$.
Let F be mid point of AC. Join F with O.
Let O is taken as origin.

Since $\vec{OD}$ is perpendicular to $\vec{AB}$

$$\begin{aligned}\vec{OD} \cdot \vec{AB} &= 0 \\ \left(\frac{\underline{a} + \underline{b}}{2}\right) \cdot (\underline{b} - \underline{a}) &= 0\end{aligned}$$

$$(\underline{b} + \underline{a}) \cdot (\underline{b} - \underline{a}) = 0 \times 2$$

$$\Rightarrow b^2 - a^2 = 0 \quad \dots\dots\dots (i)$$

Again $\vec{OE}$ is perpendicular to $\vec{BC}$

$$\begin{aligned}\vec{OE} \cdot \vec{BC} &= 0 \\ \left(\frac{\underline{b} + \underline{c}}{2}\right) \cdot (\underline{c} - \underline{b}) &= 0 \times 2\end{aligned}$$

$$(\underline{c} + \underline{b}) \cdot (\underline{c} - \underline{b}) = 0 \times 2$$

$$\Rightarrow c^2 - b^2 = 0 \quad \dots\dots\dots (ii)$$

Adding (i) & (ii) we have

$$b^2 - a^2 + c^2 - b^2 = 0$$

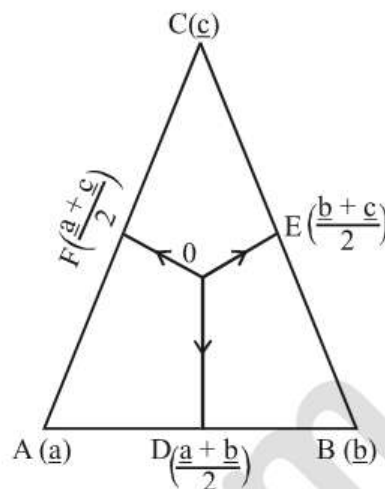
$$c^2 - a^2 = 0$$

$$(\underline{c} + \underline{a}) \cdot (\underline{c} - \underline{a}) = 0$$

$$\left(\frac{\underline{c} + \underline{a}}{2}\right) \cdot (\underline{c} - \underline{a}) = 0$$

$$\vec{OF} \cdot \vec{AC} = 0$$

Which shows that $\vec{OF}$ is perpendicular to $\vec{AC}$. Hence perpendicular bisectors of the sides of a triangle are concurrent.



Q.9 Prove that the attitudes of a triangle are concurrent. (Lahore Board 2009)

Solution:

Let AD, BE be the attitudes drawn from vertices A, B, respectively. Join C to O & produce it meet AB at F.

Since $\vec{AD} \perp \vec{BC} \Rightarrow \vec{AO} \perp \vec{BC}$ also

$$\vec{AO} \cdot \vec{BC} = 0$$

$$-\underline{a} \cdot (\underline{c} - \underline{b}) = 0$$

$$-\underline{a} \cdot \underline{c} + \underline{a} \cdot \underline{b} = 0$$

$$\underline{a} \cdot \underline{b} = \underline{a} \cdot \underline{c} \quad \dots\dots\dots (i)$$

Since $\vec{BE} \perp \vec{AC}$

$$\Rightarrow \vec{BO} \perp \vec{AC} \Rightarrow \vec{BO} \cdot \vec{AC} = 0$$

$$\Rightarrow -\underline{b} \cdot (\underline{c} - \underline{a}) = 0$$

$$\Rightarrow -\underline{b} \cdot \underline{c} + \underline{b} \cdot \underline{a} = 0$$

$$\underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{c} \quad \dots\dots\dots (ii)$$

from (i) & (ii) we have

$$\underline{a} \cdot \underline{c} = \underline{b} \cdot \underline{c}$$

$$\underline{a} \cdot \underline{c} - \underline{b} \cdot \underline{c} = 0$$

$$-\underline{c} \cdot (\underline{b} - \underline{a}) = 0$$

$$\vec{OF} \cdot \vec{AB} = 0$$

Thus $\vec{OF} \perp \vec{AB}$

$$\Rightarrow \vec{CF} \perp \vec{AB}$$

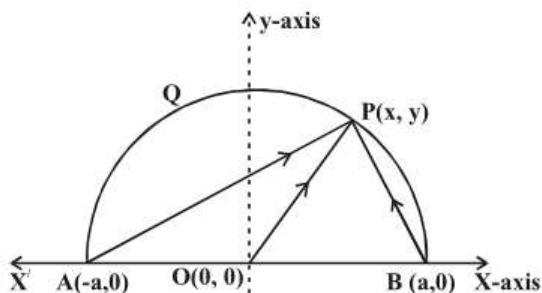
Shows attitudes of a triangle are concurrent.

Q.10 Proved that the angle in a semi circle is a right angle.

(Gujranwala Board 2006, Lahore Board, 2007)

Solution:

Let AQB be a semi circle of radius a with center at origin. Take x-axis along AB. Let $P(x,y)$ be any point on semicircle. Join A and B with P join O and P.
Now



$$\vec{OA} = -a \underline{i}$$



$$\vec{OB} = a \underline{i}$$

$$\vec{OP} = a \text{ (radius given)}$$

$$|\vec{OP}|^2 = a^2 \quad \dots\dots\dots (i)$$

$$\vec{OP} = x \underline{i} + y \underline{j}$$

$$|\vec{OP}| = \sqrt{x^2 + y^2} \Rightarrow |\vec{OP}|^2 = x^2 + y^2$$

$$= x^2 + y^2 = a^2 \quad \dots\dots\dots (ii) \quad \text{using (i)}$$

$$\vec{AP} = \vec{OP} - \vec{OA}$$

$$\vec{AP} = (x + a) \underline{i} + (y - 0) \underline{j} = (x + a) \underline{i} + y \underline{j}$$

$$\vec{BP} = \vec{OP} - \vec{OB}$$

$$= (x - a) \underline{i} + (y - 0) \underline{j} = (x - a) \underline{i} + y \underline{j}$$

$$\vec{AP} \cdot \vec{BP} = ((x + a) \underline{i} + y \underline{j}) \cdot ((x - a) \underline{i} + y \underline{j})$$

$$= x^2 - a^2 + y^2$$

$$= x^2 + y^2 - a^2$$

$$= a^2 - a^2 = 0 \quad \text{(Using ii)}$$

Hence $\vec{AP}$ is perpendicular to $\vec{BP}$.

$$\therefore \angle APB = 90^\circ$$

Q.11 Prove that $\cos(\alpha + \beta) = \cos\alpha \cos\beta - \sin\alpha \sin\beta$

(Lahore Board 2007,2011)

Solution:

Let $\hat{a}$ and $\hat{b}$ be two unit vectors making angles α and β with x-axis

Therefore, we can write

$$\hat{a} = \cos\alpha \underline{i} + \sin\alpha \underline{j}$$

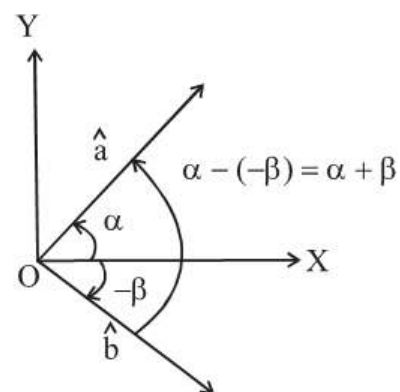
$$\hat{b} = \cos\beta \underline{i} - \sin\beta \underline{j}$$

$$\hat{a} \cdot \hat{b} = (\cos\alpha \underline{i} + \sin\alpha \underline{j}) \cdot (\cos\beta \underline{i} - \sin\beta \underline{j})$$

$$|\hat{a}| |\hat{b}| \cos(\alpha + \beta) = \cos\alpha \cos\beta - \sin\alpha \sin\beta$$

$$\cos(\alpha + \beta) = \cos\alpha \cos\beta - \sin\alpha \sin\beta$$

Hence proved



$$(\because |\hat{b}| = 1, |\hat{a}| = 1)$$



Q.12 Prove that in any triangle ABC

(i) $b = c \cos A + a \cos C$

Solution:

$$\underline{b} = \cos A + a \cos C$$

For any triangle $\underline{a} + \underline{b} + \underline{c} = 0$

$$\underline{b} = -\underline{a} - \underline{c}$$

$$\underline{b} = -(\underline{a} + \underline{c}) \dots\dots\dots (i)$$

Taking dot product with $\underline{b}$, we have

$$\underline{b} \cdot \underline{b} = -\underline{b} \cdot (\underline{a} + \underline{c})$$

$$= -\underline{b} \cdot \underline{a} - \underline{b} \cdot \underline{c}$$

$$b^2 = -|\underline{b}| |\underline{a}| \cos(\pi - C) - |\underline{b}| |\underline{c}| \cos(\pi - A)$$

$$= -ba(-\cos C) - bc(-\cos A)$$

$$b^2 = ba \cos C + bc \cos A$$

$$b = a \cos C + c \cos A$$

(Dividing throughout by b) Hence proved

(ii) $c = a \cos B + b \cos A$

Solution:

For triangle ABC, we have

$$\underline{a} + \underline{b} + \underline{c} = 0$$

$$\underline{c} = -\underline{a} - \underline{b}$$

$$\underline{c} = -(\underline{a} + \underline{b}) \dots\dots\dots (i)$$

Taking dot product with $\underline{c}$

$$\underline{c} \cdot \underline{c} = -\underline{c} \cdot (\underline{a} + \underline{b})$$

$$c^2 = -\underline{c} \cdot \underline{a} - \underline{c} \cdot \underline{b}$$

$$= -|\underline{c}| |\underline{a}| \cos(\pi - B) - |\underline{c}| |\underline{b}| \cos(\pi - A)$$

$$c^2 = -ac(-\cos B) - cb(-\cos A) \quad (\because \cos(\pi - \theta) = -\cos \theta)$$

$$c^2 = ac \cos B + bc \cos A$$

$$c = a \cos B + b \cos A \quad (\text{dividing by } c) \quad \text{Hence proved.}$$

(iii) $b^2 = c^2 + a^2 - 2ac \cos B$

Solution:

For triangle ABC, by vector addition

$$\underline{a} + \underline{b} + \underline{c} = 0$$

$$\underline{b} = -\underline{a} - \underline{c}$$

$$\underline{b} = -(\underline{a} + \underline{c}) \dots\dots\dots (i)$$

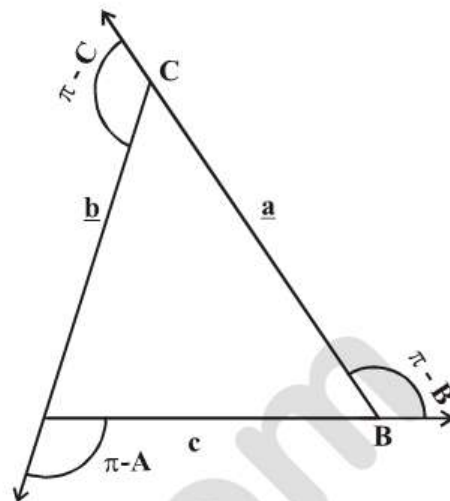
Taking dot product with $\underline{b}$

$$\underline{b} \cdot \underline{b} = -(\underline{a} + \underline{c}) \cdot \underline{b}$$

$$b^2 = -(\underline{a} + \underline{c}) \cdot -(\underline{a} + \underline{c})$$

$$= \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{c} + \underline{c} \cdot \underline{a} + \underline{c} \cdot \underline{c}$$

$$b^2 = a^2 + 2\underline{a} \cdot \underline{c} + c^2 \quad (\because \underline{a} \cdot \underline{c} = \underline{c} \cdot \underline{a})$$



$$\begin{aligned}
 b^2 &= a^2 + 2|a||c|\cos(\pi - B) + c^2 \\
 b^2 &= a^2 + 2ac(-\cos B) + c^2 \\
 b^2 &= a^2 + c^2 - 2ac\cos B
 \end{aligned}$$

Hence proved

$$(iv) \quad c^2 = a^2 + b^2 - 2ab\cos C$$

Solution:

For triangle ABC

$$\underline{a} + \underline{b} + \underline{c} = 0$$

$$\underline{c} = -\underline{a} - \underline{b}$$

$$\underline{c} = -(\underline{a} + \underline{b}) \quad \dots\dots (i)$$

Taking dot product by $\underline{c}$

$$\underline{c} \cdot \underline{c} = -(\underline{a} + \underline{b}) \cdot \underline{c}$$

$$c^2 = -(\underline{a} + \underline{b}) \cdot -(\underline{a} + \underline{b})$$

$$= \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{a} + \underline{b} \cdot \underline{b}$$

$$c^2 = a^2 + 2\underline{a} \cdot \underline{b} + b^2 \quad (\because \underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{a})$$

$$c^2 = a^2 + 2|a||b|\cos(\pi - C) + b^2$$

$$c^2 = a^2 + 2ab(-\cos C) + b^2$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

Hence proved

The Cross Product or Vector Product of Two Vectors

Let $\underline{u}$ & $\underline{v}$ be two vectors. The cross or vector product of $\underline{u}$ and $\underline{v}$ is defined as

$$\underline{u} \times \underline{v} = |\underline{u}| |\underline{v}| \sin\theta \hat{n}$$

When $\hat{n}$ is unit vector perpendicular to the plane of $\underline{u}$ and $\underline{v}$.

$$\underline{u} \times \underline{v} = |\underline{u}| |\underline{v}| \sin\theta \hat{n}$$

Where

$$\hat{n} = \frac{\underline{u} \times \underline{v}}{|\underline{u} \times \underline{v}|}$$

$$\sin\theta = \frac{|\underline{u} \times \underline{v}|}{|\underline{u}| |\underline{v}| \hat{n}}$$



$$\sin\theta = \frac{|\underline{u} \times \underline{v}|}{|\underline{u}| |\underline{v}|}$$

Important Points;

- (i) $\underline{i} \times \underline{i} = \underline{j} \times \underline{j} = \underline{k} \times \underline{k} = 0$
- (ii) $\underline{i} \times \underline{j} = \underline{k}$, $\underline{j} \times \underline{k} = \underline{i}$, $\underline{k} \times \underline{i} = \underline{j}$
- (iii) $\underline{i} \times \underline{j} \neq \underline{j} \times \underline{i}$ i.e., Cross product is not commutative
- (vi) Area of parallelogram = $|\underline{u} \times \underline{v}|$
- (v) Area of triangle = $\frac{1}{2} |\underline{u} \times \underline{v}|$

Parallel vectors:

If $\underline{u}$ & $\underline{v}$ are parallel vectors then $\underline{u} \times \underline{v} = 0$



EXERCISE 7.4

Q.1 Compute the cross product $\underline{a} \times \underline{b}$ and $\underline{b} \times \underline{a}$, check your answer by showing that each $\underline{a}$ and $\underline{b}$ is perpendicular to $\underline{a} \times \underline{b}$ and $\underline{b} \times \underline{a}$.

(i) $\underline{a} = 2\underline{i} + \underline{j} - \underline{k}$, $\underline{b} = \underline{i} - \underline{j} + \underline{k}$

Solution:

$$\begin{aligned}\underline{a} \times \underline{b} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{vmatrix} \\ &= \underline{i} \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} - \underline{j} \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} + \underline{k} \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} \\ &= \underline{i} (1 - 1) - \underline{j} (2 + 1) + \underline{k} (-2 - 1) \\ \underline{a} \times \underline{b} &= 0\underline{i} - 3\underline{j} - 3\underline{k}\end{aligned}$$

We will show that $\underline{a}$ is perpendicular to $\underline{a} \times \underline{b}$, for this we have $\underline{a} \cdot (\underline{a} \times \underline{b})$

$$\begin{aligned}&= (2\underline{i} + \underline{j} - \underline{k}) \cdot (0\underline{i} - 3\underline{j} - 3\underline{k}) \\ &= 0 - 3 + 3 = 0\end{aligned}$$

$\therefore$ $\underline{a}$ and $\underline{a} \times \underline{b}$ are perpendicular.

Next, we will show that $\underline{b}$ is perpendicular to $\underline{a} \times \underline{b}$. For this we have $\underline{b} \cdot (\underline{a} \times \underline{b})$

$$(\underline{i} - \underline{j} + \underline{k}) \cdot (0\underline{i} - 3\underline{j} - 3\underline{k})$$



$$= 0 + 3 - 3 = 0$$

Hence $\underline{b}$ is perpendicular to $\underline{a} \times \underline{b}$.

$$\begin{aligned}\underline{b} \times \underline{a} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix} \\ &= \underline{i} \begin{vmatrix} -1 & 1 \\ 1 & -1 \end{vmatrix} - \underline{j} \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} + \underline{k} \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} \\ &= \underline{i} (1 - 1) - \underline{j} (-1 - 2) + \underline{k} (1 + 2) \\ \underline{b} \times \underline{a} &= 0\underline{i} + 3\underline{j} + 3\underline{k}\end{aligned}$$

We will show that $\underline{a}$ is perpendicular to $\underline{b} \times \underline{a}$.

$$\begin{aligned}\underline{a} \cdot (\underline{b} \times \underline{a}) &= (2\underline{i} + \underline{j} - \underline{k}) \cdot (0\underline{i} + 3\underline{j} + 3\underline{k}) \\ &= 0 + 3 - 3 \\ &= 0\end{aligned}$$

Hence $\underline{a}$ and $\underline{b} \times \underline{a}$ are perpendicular to each other

Next, we will show that $\underline{b}$ is perpendicular to $\underline{b} \times \underline{a}$

$$\begin{aligned}\underline{b} \cdot (\underline{b} \times \underline{a}) &= (\underline{i} - \underline{j} + \underline{k}) \cdot (0\underline{i} + 3\underline{j} + 3\underline{k}) \\ &= 0 - 3 + 3 = 0\end{aligned}$$

Hence $\underline{b}$ and $\underline{b} \times \underline{a}$ are perpendicular to each other.

$$(ii) \quad \underline{a} = \underline{i} + \underline{j} + 0\underline{k}, \quad \underline{b} = \underline{i} - \underline{j} + 0\underline{k}$$

(Lahore Board 2009)

Solution:

$$\begin{aligned}\underline{a} \times \underline{b} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix} \\ &= \underline{i} \begin{vmatrix} 1 & 0 \\ -1 & 0 \end{vmatrix} - \underline{j} \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} + \underline{k} \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} \\ &= 0\underline{i} - 0\underline{j} + \underline{k} (-1 - 1) \\ &= -2\underline{k}\end{aligned}$$

We will show that $\underline{a}$ is perpendicular to $\underline{a} \times \underline{b}$.

$$\begin{aligned}\text{For this } \underline{a} \cdot (\underline{a} \times \underline{b}) &= (\underline{i} + \underline{j} + 0\underline{k}) \cdot (0\underline{i} + 0\underline{j} - 2\underline{k}) \\ &= 0 + 0 + 0 = 0\end{aligned}$$

Next, we will show that $\underline{b}$ is perpendicular to $\underline{a} \times \underline{b}$

$$\begin{aligned}\text{For this } \underline{b} \cdot (\underline{a} \times \underline{b}) &= (\underline{i} - \underline{j} + 0\underline{k}) \cdot (0\underline{i} + 0\underline{j} - 2\underline{k})\end{aligned}$$



$$= 0 + 0 + 0 = 0$$

Next, we will show that $\underline{b}$ is perpendicular to $\underline{a} \times \underline{b}$

For this

$$\begin{aligned} \underline{b} \cdot (\underline{a} \times \underline{b}) &= (\underline{i} - \underline{j} + 0\underline{k}) \cdot (0\underline{i} + 0\underline{j} - 2\underline{k}) \\ &= 0 + 0 + 0 = 0 \end{aligned}$$

Hence

$$\begin{aligned} \underline{b} \times \underline{a} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & -1 & 0 \\ 1 & 1 & 0 \end{vmatrix} \\ &= \underline{i} \begin{vmatrix} -1 & 0 \\ 1 & 0 \end{vmatrix} - \underline{j} \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} + \underline{k} \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} \\ &= 0\underline{i} - 0\underline{j} + \underline{k} (1 + 1) \\ \underline{b} \times \underline{a} &= 0\underline{i} - 0\underline{j} + 2\underline{k} \end{aligned}$$

We will show that $\underline{a}$ is perpendicular to $\underline{b} \times \underline{a}$.

For this

$$\begin{aligned} \underline{a} \cdot (\underline{b} \times \underline{a}) &= (\underline{i} + \underline{j} + 0\underline{k}) \cdot (0\underline{i} + 0\underline{j} + 2\underline{k}) \\ &= 0 + 0 + 0 \\ &= 0 \end{aligned}$$

Hence $\underline{a}$ and $\underline{b} \times \underline{a}$ are perpendicular to each other. Next, we will show that $\underline{b}$ is perpendicular to $\underline{b} \times \underline{a}$

For this

$$\begin{aligned} \underline{b} \cdot (\underline{b} \times \underline{a}) &= (\underline{i} - \underline{j} + 0\underline{k}) \cdot (0\underline{i} + 0\underline{j} + 2\underline{k}) \\ &= 0 + 0 + 0 \\ &= 0 \end{aligned}$$

Hence $\underline{b}$ and $\underline{b} \times \underline{a}$ are perpendicular to each other.

$$(iii) \quad \underline{a} = 3\underline{i} - 2\underline{j} + \underline{k}, \quad \underline{b} = \underline{i} + \underline{j} + 0\underline{k}$$

Solution:

$$\begin{aligned} \underline{a} \times \underline{b} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 3 & -2 & 1 \\ 1 & 1 & 0 \end{vmatrix} \\ &= \underline{i} \begin{vmatrix} -2 & 1 \\ 1 & 0 \end{vmatrix} - \underline{j} \begin{vmatrix} 3 & 1 \\ 1 & 0 \end{vmatrix} + \underline{k} \begin{vmatrix} 3 & -2 \\ 1 & 1 \end{vmatrix} \\ &= \underline{i} (0 - 1) - \underline{j} (0 - 1) + \underline{k} (3 + 2) \end{aligned}$$



$$\underline{a} \times \underline{b} = -\underline{i} + \underline{j} + 5\underline{k}$$

We will show that $\underline{a}$ is perpendicular to $\underline{a} \times \underline{b}$.

For this $\underline{a} \cdot (\underline{a} \times \underline{b})$

$$= (3\underline{i} - 2\underline{j} + \underline{k}) \cdot (-\underline{i} + \underline{j} + 5\underline{k})$$

$$= -3 - 2 + 5 = 0$$

$\therefore$ $\underline{a}$ and $\underline{a} \times \underline{b}$ are perpendicular to each other.

Next $\underline{b} \cdot (\underline{a} \times \underline{b})$

$$= (\underline{i} + \underline{j} + 0\underline{k}) \cdot (-\underline{i} + \underline{j} + 5\underline{k})$$

$$= -1 + 1 + 0 = 0$$

Hence $\underline{b}$ and $\underline{a} \times \underline{b}$ are perpendicular to each other.

$$\underline{b} \times \underline{a} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 1 & 0 \\ 3 & -2 & 1 \end{vmatrix}$$

$$= \underline{i} \begin{vmatrix} 1 & 0 \\ -2 & 1 \end{vmatrix} - \underline{j} \begin{vmatrix} 1 & 0 \\ 3 & 1 \end{vmatrix} + \underline{k} \begin{vmatrix} 1 & 1 \\ 3 & -2 \end{vmatrix}$$

$$= \underline{i}(1 - 0) - \underline{j}(1 - 0) + \underline{k}(-2 - 3)$$

$$\underline{b} \times \underline{a} = \underline{i} - \underline{j} - 5\underline{k}$$

We will show that $\underline{a}$ and $\underline{b} \times \underline{a}$ are perpendicular to each other.

For this $\underline{a} \cdot (\underline{b} \times \underline{a})$

$$= (3\underline{i} - 2\underline{j} + \underline{k}) \cdot (\underline{i} - \underline{j} - 5\underline{k})$$

$$= 3 + 2 - 5 = 0$$

$\therefore$ $\underline{a}$ and $\underline{b} \times \underline{a}$ are perpendicular to each other.

Next,

We will show that $\underline{b}$ is perpendicular to $\underline{b} \times \underline{a}$

For this

$\underline{b} \cdot (\underline{b} \times \underline{a})$

$$= (\underline{i} + \underline{j} + 0\underline{k}) \cdot (\underline{i} - \underline{j} - 5\underline{k})$$

$$= 1 - 1 + 0 = 0$$

Hence $\underline{b}$ is perpendicular to $\underline{b} \times \underline{a}$

$$(iv) \quad \underline{a} = -4\underline{i} + \underline{j} - 2\underline{k}, \quad \underline{b} = 2\underline{i} + \underline{j} + \underline{k}$$

Solution:

$$\underline{a} \times \underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -4 & 1 & -2 \\ 2 & 1 & 1 \end{vmatrix}$$



$$\begin{aligned}
 &= \underline{i} \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix} - \underline{j} \begin{vmatrix} -4 & -2 \\ 2 & 1 \end{vmatrix} + \underline{k} \begin{vmatrix} -4 & 1 \\ 2 & 1 \end{vmatrix} \\
 &= \underline{i} (1 + 2) - \underline{j} (-4 + 4) + \underline{k} (-4 - 2) \\
 \underline{a} \times \underline{b} &= 3\underline{i} + 0\underline{j} - 6\underline{k}
 \end{aligned}$$

We will show that $\underline{a}$ is perpendicular to $\underline{a} \times \underline{b}$.

For this $\underline{a} \cdot (\underline{a} \times \underline{b})$

$$\begin{aligned}
 &= (-4\underline{i} + \underline{j} - 2\underline{k}) \cdot (3\underline{i} + 0\underline{j} - 6\underline{k}) \\
 &= -12 + 0 + 12 = 0
 \end{aligned}$$

Hence $\underline{a}$ is perpendicular to $\underline{a} \times \underline{b}$.

Next,

We will show that $\underline{b}$ is perpendicular to $\underline{a} \times \underline{b}$.

For this

$$\begin{aligned}
 &\underline{b} \cdot (\underline{a} \times \underline{b}) \\
 \underline{b} \cdot (\underline{a} \times \underline{b}) &= (2\underline{i} + \underline{j} + \underline{k}) \cdot (3\underline{i} + 0\underline{j} - 6\underline{k}) \\
 &= 6 + 0 - 6 = 0
 \end{aligned}$$

Hence $\underline{b}$ is perpendicular to $\underline{a} \times \underline{b}$.

Now

$$\begin{aligned}
 \underline{b} \times \underline{a} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & 1 & 1 \\ -4 & 1 & -2 \end{vmatrix} \\
 &= \underline{i} (-2 - 1) - \underline{j} (-4 + 4) + \underline{k} (2 + 4) \\
 &= -3\underline{i} + 0\underline{j} + 6\underline{k}
 \end{aligned}$$

We will show that $\underline{a}$ is perpendicular to $\underline{b} \times \underline{a}$.

For this $\underline{a} \cdot (\underline{b} \times \underline{a})$

$$\begin{aligned}
 &= (-4\underline{i} + \underline{j} - 2\underline{k}) \cdot (-3\underline{i} + 0\underline{j} + 6\underline{k}) \\
 &= 12 + 0 - 12 = 0
 \end{aligned}$$

Hence $\underline{a}$ is perpendicular to $\underline{b} \times \underline{a}$.

Next,

We will show that $\underline{b}$ is perpendicular to $\underline{b} \times \underline{a}$.

$$\underline{b} \cdot (\underline{b} \times \underline{a}) = (2\underline{i} + \underline{j} + \underline{k}) \cdot (-3\underline{i} + 0\underline{j} + 6\underline{k})$$



$$= -6 + 6 = 0$$

Hence $\underline{b}$ & $\underline{b} \times \underline{a}$ are perpendicular to each other.

Q.2 Find the unit vector perpendicular to the plane containing $\underline{a}$ & $\underline{b}$. Also find Sine of angle between them.

(i) $\underline{a} = 2\underline{i} - 6\underline{j} - 3\underline{k}$ $\underline{b} = 4\underline{i} + 3\underline{j} - \underline{k}$

(Lahore Board 2009)

Solution:

$$\underline{a} \times \underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & -6 & -3 \\ 4 & 3 & -1 \end{vmatrix}$$

$$= \underline{i}(6+9) - \underline{j}(-2+12) + \underline{k}(6+24)$$

$$\underline{a} \times \underline{b} = 15\underline{i} - 10\underline{j} + 30\underline{k}$$

$$|\underline{a} \times \underline{b}| = \sqrt{(15)^2 + (-10)^2 + (30)^2} = \sqrt{225 + 100 + 900} = \sqrt{1225}$$

$$|\underline{a} \times \underline{b}| = 35$$

$$\text{Required unit vector} = \frac{\underline{a} \times \underline{b}}{|\underline{a} \times \underline{b}|} = \frac{15\underline{i} - 10\underline{j} + 30\underline{k}}{35}$$

$$= \frac{15}{35}\underline{i} - \frac{10}{35}\underline{j} + \frac{30}{35}\underline{k}$$

$$= \frac{3}{7}\underline{i} - \frac{2}{7}\underline{j} + \frac{6}{7}\underline{k}$$

$$|\underline{a}| = \sqrt{(2)^2 + (-6)^2 + (-3)^2} = \sqrt{4 + 36 + 9} = \sqrt{49} = 7$$

$$|\underline{b}| = \sqrt{(4)^2 + (3)^2 + (-1)^2} = \sqrt{16 + 9 + 1} = \sqrt{26}$$

$$\sin\theta = \frac{|\underline{a} \times \underline{b}|}{|\underline{a}| |\underline{b}|} = \frac{35}{7\sqrt{26}}$$

$$\sin\theta = \frac{5}{\sqrt{26}} \quad \text{Ans.}$$

(ii) $\underline{a} = -\underline{i} - \underline{j} - \underline{k}$, $\underline{b} = 2\underline{i} - 3\underline{j} + 4\underline{k}$

Solution:

$$\underline{a} \times \underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -1 & -1 & -1 \\ 2 & -3 & 4 \end{vmatrix}$$

$$= \underline{i}(-4-3) - \underline{j}(-4+2) + \underline{k}(3+2)$$

$$\underline{a} \times \underline{b} = -7\underline{i} + 2\underline{j} + 5\underline{k}$$

$$|\underline{a} \times \underline{b}| = \sqrt{(-7)^2 + (2)^2 + (5)^2} = \sqrt{49 + 4 + 25} = \sqrt{78}$$



required unit vector

$$\begin{aligned}\hat{n} &= \frac{\underline{a} \times \underline{b}}{|\underline{a} \times \underline{b}|} \\ \hat{n} &= \frac{-7\underline{i} + 2\underline{j} + 5\underline{k}}{\sqrt{78}} \\ &= \frac{-7}{\sqrt{78}}\underline{i} + \frac{2}{\sqrt{78}}\underline{j} + \frac{5}{\sqrt{78}}\underline{k}\end{aligned}$$

$$\begin{aligned}|\underline{a}| &= \sqrt{(-1)^2 + (-1)^2 + (-1)^2} \\ &= \sqrt{1+1+1} = \sqrt{3}\end{aligned}$$

$$|\underline{b}| = \sqrt{(2)^2 + (-3)^2 + (4)^2} = \sqrt{4+9+16} = \sqrt{29}$$

$$\begin{aligned}\sin\theta &= \frac{|\underline{a} \times \underline{b}|}{|\underline{a}| |\underline{b}|} = \frac{\sqrt{78}}{\sqrt{3} \sqrt{29}} \\ &= \frac{\sqrt{3 \times 78}}{\sqrt{3} \times \sqrt{29}} = \sqrt{\frac{26}{29}} \quad \text{Ans}\end{aligned}$$

(iii) $\underline{a} = 2\underline{i} - 2\underline{j} + 4\underline{k}$, $\underline{b} = -\underline{i} + \underline{j} - 2\underline{k}$

Solution:

$$\begin{aligned}\underline{a} \times \underline{b} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & -2 & 4 \\ -1 & 1 & -2 \end{vmatrix} \\ &= \underline{i} \begin{vmatrix} -2 & 4 \\ 1 & -2 \end{vmatrix} - \underline{j} \begin{vmatrix} 2 & 4 \\ -1 & -2 \end{vmatrix} + \underline{k} \begin{vmatrix} 2 & -2 \\ -1 & 1 \end{vmatrix} \\ &= \underline{i} (4 - 4) - \underline{j} (-4 + 4) + \underline{k} (2 - 2) \\ &= 0\underline{i} + 0\underline{j} + 0\underline{k} \\ |\underline{a}| &= \sqrt{(2)^2 + (-2)^2 + (4)^2} = \sqrt{4+4+16} = \sqrt{24} \\ |\underline{b}| &= \sqrt{(-1)^2 + (1)^2 + (-2)^2} = \sqrt{1+1+4} = \sqrt{6} \\ |\underline{a} \times \underline{b}| &= \sqrt{0} = 0\end{aligned}$$

It is not possible to find out the required unit vector.

$$\sin\theta = \frac{|\underline{a} \times \underline{b}|}{|\underline{a}| |\underline{b}|}$$



$$\begin{aligned} &= \frac{0}{\sqrt{24} \sqrt{6}} \\ \sin \theta &= 0 \quad \text{Ans.} \end{aligned}$$

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$$(iv) \quad \underline{a} = \underline{i} + \underline{j}, \quad \underline{b} = \underline{i} - \underline{j}$$

Solution:

$$\begin{aligned} \underline{a} \times \underline{b} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix} \\ &= \underline{i}(0-0) - \underline{j}(0-0) + \underline{k}(-1-1) \\ \underline{a} \times \underline{b} &= 0\underline{i} - 0\underline{j} - 2\underline{k} \\ |\underline{a} \times \underline{b}| &= \sqrt{(0)^2 + (0)^2 + (-2)^2} = \sqrt{4} = 2 \end{aligned}$$

Required unit vector

$$\begin{aligned} \hat{n} &= \frac{|\underline{a} \times \underline{b}|}{|\underline{a}| |\underline{b}|} = \frac{-2\underline{k}}{2} = -\underline{k} \\ |\underline{a}| &= \sqrt{(-1)^2 + (1)^2} = \sqrt{2} \\ |\underline{b}| &= \sqrt{1+1} = \sqrt{2} \\ \sin\theta &= \frac{|\underline{a} \times \underline{b}|}{|\underline{a}| |\underline{b}|} = \frac{2}{\sqrt{2}\sqrt{2}} = \frac{2}{2} \\ \sin\theta &= 1 \quad \text{Ans.} \end{aligned}$$

Q.3 Find the area of triangle, determined by the point P, Q and R.

$$(i) \quad P(0, 0, 0); \quad Q(2, 3, 2); \quad R(-1, 1, 4)$$

Solution:

$$P(0, 0, 0); \quad Q(2, 3, 2); \quad R(-1, 1, 4)$$

$$\text{Area of triangle having P, Q, R as its vertices} = \frac{1}{2} |\vec{PQ} \times \vec{PR}|$$

$$\begin{aligned} \vec{PQ} &= \vec{OQ} - \vec{OP} \\ &= (2-0)\underline{i} + (3-0)\underline{j} + (2-0)\underline{k} \end{aligned}$$

$$\vec{PQ} = 2\underline{i} + 3\underline{j} + 2\underline{k}$$

$$\begin{aligned} \vec{PR} &= \vec{OR} - \vec{OP} \\ &= (-1-0)\underline{i} + (1-0)\underline{j} + (4-0)\underline{k} \end{aligned}$$

$$\vec{PR} = -\underline{i} - \underline{j} + 4\underline{k}$$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & 3 & 2 \\ -1 & -1 & 4 \end{vmatrix}$$



$$= \underline{i} (12 - 2) - \underline{j} (8 + 2) + \underline{k} (2 + 3)$$

$$= 10\underline{i} - 10\underline{j} + 5\underline{k}$$

$$|\vec{PQ} \times \vec{PR}| = \sqrt{(10)^2 + (-10)^2 + (5)^2} = \sqrt{100 + 100 + 25} = \sqrt{225} = 15$$

$$\begin{aligned} \text{Area of triangle} &= \frac{1}{2} |\vec{PQ} \times \vec{PR}| \\ &= \frac{1}{2} (15) \\ &= \frac{15}{2} \text{ sq. units} \quad \text{Ans.} \end{aligned}$$

(ii) $P(1, -1, -1); Q(2, 0, -1); R(0, 2, 1)$

Solution:

$$\begin{aligned} \vec{PQ} &= \vec{OQ} - \vec{OP} \\ &= (2 - 1)\underline{i} + (0 + 1)\underline{j} + (-1 + 1)\underline{k} \end{aligned}$$

$$\vec{PQ} = \underline{i} + \underline{j} + 0\underline{k}$$

$$\begin{aligned} \vec{PR} &= \vec{OR} - \vec{OP} \\ &= (0 - 1)\underline{i} + (2 - 1)\underline{j} + (1 + 1)\underline{k} \\ &= -\underline{i} + \underline{j} + 2\underline{k} \end{aligned}$$

$$\begin{aligned} \vec{PQ} \times \vec{PR} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 1 & 0 \\ -1 & 1 & 2 \end{vmatrix} \\ &= \underline{i} (2 - 0) - \underline{j} (2 - 0) + \underline{k} (3 + 1) \\ &= 2\underline{i} - 2\underline{j} + 4\underline{k} \end{aligned}$$

$$|\vec{PQ} \times \vec{PR}| = \sqrt{(2)^2 + (-2)^2 + (4)^2} = \sqrt{4 + 4 + 16} = \sqrt{24}$$

$$\text{Area of triangle} = \frac{1}{2} |\vec{PQ} \times \vec{PR}| = \frac{1}{2} \sqrt{24} = \frac{1}{2} (2\sqrt{6}) = \sqrt{6} \text{ sq. units} \quad \text{Ans.}$$

Q.4 Find the area of parallelogram, whose vertices are

(i) $A(0, 0, 0), B(1, 2, 3); C(2, -1, 1); D(3, 1, 4)$

Solution:

$A(0, 0, 0), B(1, 2, 3); C(2, -1, 1); D(3, 1, 4)$

$$\text{Area of parallelogram } |\vec{AB} \times \vec{AC}|$$



$$\begin{aligned}
 \vec{AB} &= \vec{OB} - \vec{OA} \\
 &= \underline{i} + 2\underline{j} + 3\underline{k} \\
 \vec{AC} &= \vec{OC} - \vec{OA} \\
 &= 2\underline{i} - \underline{j} + \underline{k} \\
 \vec{AB} \times \vec{AC} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 2 & 3 \\ 2 & -1 & 1 \end{vmatrix} \\
 &= \underline{i}(2+3) - \underline{j}(1-6) + \underline{k}(-1-4) \\
 &= 5\underline{i} + 5\underline{j} - 5\underline{k}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of parallelogram} &= |\vec{AB} \times \vec{AC}| \\
 &= \sqrt{(5)^2 + (5)^2 + (-5)^2} = \sqrt{25 + 25 + 25} = \sqrt{75} \text{ sq. units} \\
 &= 5\sqrt{3} \text{ sq. units} \quad \text{Ans.}
 \end{aligned}$$

(ii) A (1, 2, -1); B (4, 2, -3); C (6, -5, 2); D (9, -5, 0)

Solution:

$$\begin{aligned}
 \vec{AB} &= \vec{OB} - \vec{OA} \\
 &= (4-1)\underline{i} + (2-2)\underline{j} + (-3+1)\underline{k} \\
 &= 3\underline{i} + 0\underline{j} - 2\underline{k} \\
 \vec{AC} &= \vec{OC} - \vec{OA} \\
 &= (6-1)\underline{i} + (-5-2)\underline{j} + (2+1)\underline{k} \\
 &= 5\underline{i} - 7\underline{j} + 3\underline{k} \\
 \vec{AB} \times \vec{AC} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 3 & 0 & -2 \\ 5 & -7 & 3 \end{vmatrix} \\
 &= \underline{i}(0-14) - \underline{j}(9+10) + \underline{k}(-21+0) \\
 &= -14\underline{i} - 19\underline{j} - 21\underline{k}
 \end{aligned}$$

$$|\vec{AB} \times \vec{AC}| = \sqrt{(-14)^2 + (-19)^2 + (-21)^2} = \sqrt{196 + 361 + 441} = \sqrt{998}$$

$$\text{Area of parallelogram} = |\vec{AB} \times \vec{AC}| = \sqrt{998} \text{ sq. units} \quad \text{Ans.}$$



(iii) $A(-1, 1, 1); B(-1, 2, 2); C(-3, 4, -5); D(-3, 5, -4)$

Solution:

$$A(-1, 1, 1); B(-1, 2, 2); C(-3, 4, -5); D(-3, 5, -4)$$

$$\begin{aligned}\vec{AB} &= \vec{OB} - \vec{OA} \\ &= (-1+1)\underline{i} + (2-1)\underline{j} + (2-1)\underline{k}\end{aligned}$$

$$\vec{AB} = 0\underline{i} + \underline{j} + \underline{k}$$

$$\begin{aligned}\vec{AC} &= \vec{OC} - \vec{OA} \\ &= (-3+1)\underline{i} + (4-1)\underline{j} + (-5-1)\underline{k} \\ &= -2\underline{i} + 3\underline{j} - 6\underline{k}\end{aligned}$$

$$\begin{aligned}\vec{AB} \times \vec{AC} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & 1 & 1 \\ -2 & 3 & -6 \end{vmatrix} \\ &= \underline{i}(-6-3) - \underline{j}(0+2) + \underline{k}(0+2) \\ &= -9\underline{i} - 2\underline{j} + 2\underline{k}\end{aligned}$$

$$|\vec{AB} \times \vec{AC}| = \sqrt{(-9)^2 + (-2)^2 + (2)^2} = \sqrt{81+4+4} = \sqrt{89} \text{ sq. units}$$

$$\text{Area of parallelogram} = |\vec{AB} \times \vec{AC}| = \sqrt{89} \text{ sq. units} \quad \text{Ans.}$$

Q.5 Which vectors if any, are perpendicular or parallel.

$$(i) \quad \underline{u} = 5\underline{i} - \underline{j} + \underline{k}, \quad \underline{v} = 0\underline{i} + \underline{j} - 5\underline{k}; \quad \underline{w} = -15\underline{i} + 3\underline{j} - 3\underline{k}$$

Solution:

$$\begin{aligned}\underline{u} \cdot \underline{v} &= (5\underline{i} - \underline{j} + \underline{k}) \cdot (0\underline{i} + \underline{j} - 5\underline{k}) \\ &= 0 - 1 - 5 = -6 \neq 0\end{aligned}$$

So $\underline{u}$ & $\underline{v}$ are not perpendicular to each other.

$$\begin{aligned}\underline{u} \times \underline{v} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 5 & -1 & 1 \\ 0 & 1 & -5 \end{vmatrix} \\ &= \underline{i}(5-1) - \underline{j}(-25-0) + \underline{k}(5-0) \\ &= 4\underline{i} + 25\underline{j} + 5\underline{k} \\ &\neq 0\end{aligned}$$

So $\underline{u}$ and $\underline{v}$ are not parallel

$$\underline{w} = -15\underline{i} + 3\underline{j} - 3\underline{k}$$



$$\begin{aligned}
 \underline{w} &= -3(5\underline{i} - \underline{j} + \underline{k}) \\
 \underline{w} &= -3\underline{u} \Rightarrow \underline{w} = \lambda \underline{u}, \lambda \in \mathbb{R} \text{ Hence } \underline{u} \& \underline{w} \text{ are parallel} \\
 \underline{v} \cdot \underline{w} &= (0\underline{i} + \underline{j} - 5\underline{k}) \cdot (-15\underline{i} + 3\underline{j} - 3\underline{k}) \\
 &= 0 + 3 + 15 = 18 \neq 0
 \end{aligned}$$

Hence $\underline{v}$ & $\underline{w}$ are not perpendicular.

$\underline{v}$ & $\underline{w}$ cannot be written $\underline{v} = \lambda \underline{w}$, $\lambda \in \mathbb{R}$ so they are not parallel.

$$(ii) \quad \underline{u} = \underline{i} + 2\underline{j} - \underline{k}; \quad \underline{v} = -\underline{i} + \underline{j} + \underline{k}; \quad \underline{w} = \frac{-\pi}{2}\underline{i} - \pi\underline{j} + \frac{\pi}{2}\underline{k}$$

Solution:

$$\begin{aligned}
 \underline{u} \cdot \underline{v} &= (\underline{i} + 2\underline{j} - \underline{k}) \cdot (-\underline{i} + \underline{j} + \underline{k}) \\
 &= -1 + 2 - 1 = 0
 \end{aligned}$$

Therefore $\underline{u}$ and $\underline{v}$ are perpendicular to each other.

$$\begin{aligned}
 \underline{w} &= -\frac{\pi}{2}\underline{i} - \pi\underline{j} + \frac{\pi}{2}\underline{k} \\
 &= \frac{-\pi\underline{i} - 2\pi\underline{j} + \pi\underline{k}}{2} \\
 &= \frac{1}{2} [-\pi\underline{i} - 2\pi\underline{j} + \pi\underline{k}] \\
 &= \frac{-\pi}{2} [\underline{i} + 2\underline{j} - \underline{k}]
 \end{aligned}$$

$$\underline{w} = -\frac{\pi}{2}\underline{u} \Rightarrow \underline{w} = \lambda \underline{u}, \lambda \in \mathbb{R}$$

Hence $\underline{u}$ & $\underline{w}$ are parallel

$$\begin{aligned}
 \underline{v} \cdot \underline{w} &= (-\underline{i} + \underline{j} + \underline{k}) \cdot \left(-\frac{\pi}{2}\underline{i} - \pi\underline{j} + \frac{\pi}{2}\underline{k}\right) \\
 &= \frac{\pi}{2} - \pi + \frac{\pi}{2} \\
 &= \pi - \pi = 0
 \end{aligned}$$

$\therefore$ $\underline{v}$ & $\underline{w}$ are perpendicular

Q.6 Prove that $\underline{a} \times (\underline{b} + \underline{c}) + \underline{b} \times (\underline{c} + \underline{a}) + \underline{c} \times (\underline{a} + \underline{b}) = 0$

(Lahore Board 2005)

Solution:

$$\begin{aligned}
 \text{L.H.S} \quad & \underline{a} \times (\underline{b} + \underline{c}) + \underline{b} \times (\underline{c} + \underline{a}) + \underline{c} \times (\underline{a} + \underline{b}) \\
 &= \underline{a} \times \underline{b} + \underline{a} \times \underline{c} + \underline{b} \times \underline{c} + \underline{b} \times \underline{a} + \underline{c} \times \underline{a} + \underline{c} \times \underline{b} \\
 &= \underline{a} \times \underline{b} + \underline{a} \times \underline{c} + \underline{b} \times \underline{c} - \underline{a} \times \underline{b} - \underline{a} \times \underline{c} - \underline{b} \times \underline{c} \\
 &= 0 = \text{R.H.S} \quad \text{Hence proved.}
 \end{aligned}$$

Hence proved



Q.7 If $\underline{a} + \underline{b} + \underline{c} = \underline{0}$, then prove that

$$\underline{a} \times \underline{b} = \underline{b} \times \underline{c} = \underline{c} \times \underline{a}$$

(Gujranwala Board 2005)

Solution:

$$\underline{a} + \underline{b} + \underline{c} = \underline{0}$$

$$\underline{a} = -\underline{b} - \underline{c}$$

$$\underline{a} = -(\underline{b} + \underline{c}) \quad \text{Taking cross product with } \underline{b}$$

$$\underline{a} \times \underline{b} = -(\underline{b} + \underline{c}) \times \underline{b}$$

$$\underline{a} \times \underline{b} = -\underline{b} \times \underline{b} - \underline{c} \times \underline{b}$$

$$\underline{a} \times \underline{b} = \underline{0} - \underline{c} \times \underline{b}$$

$$\underline{a} \times \underline{b} = \underline{b} \times \underline{c} \quad \dots\dots (i)$$

Again

$$\underline{a} + \underline{b} + \underline{c} = \underline{0}$$

$$\underline{b} = -\underline{a} - \underline{c}$$

Taking cross product with $\underline{c}$

$$\underline{b} \times \underline{c} = -(\underline{a} + \underline{c}) \times \underline{c}$$

$$= -\underline{a} \times \underline{c} - \underline{c} \times \underline{c}$$

$$\underline{b} \times \underline{c} = \underline{c} \times \underline{a} \quad \dots\dots (ii)$$

from (i) & (ii) we have

$$\underline{a} \times \underline{b} = \underline{b} \times \underline{c} = \underline{c} \times \underline{a}$$

Hence proved

Q.8 Proved that $\sin(\alpha - \beta) = \sin\alpha \cos\beta - \cos\alpha \sin\beta$

(Gujranwala Board 2003, Lahore Board, 2009)

Solution:

Let $\hat{a}$, $\hat{b}$ be two unit vectors making angles α , β with x-axis respectively.

$$\hat{a} = \cos\alpha \underline{i} + \sin\alpha \underline{j} + 0\underline{k}$$

$$\hat{b} = \cos\beta \underline{i} + \sin\beta \underline{j} + 0\underline{k}$$

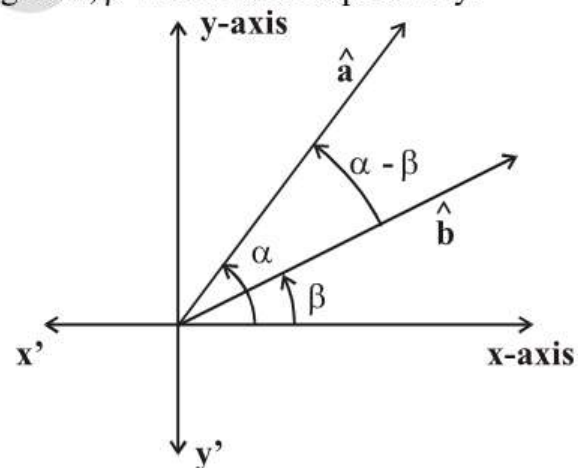
$$\hat{b} \times \hat{a} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \cos\beta & \sin\beta & 0 \\ \cos\alpha & \sin\alpha & 0 \end{vmatrix}$$

$$= \underline{i} (0-0) - \underline{j} (0-0) + \underline{k} (\sin\alpha \cos\beta - \cos\alpha \sin\beta)$$

$$\hat{b} \times \hat{a} = \underline{k} (\sin\alpha \cos\beta - \cos\alpha \sin\beta)$$

$$|\hat{b}| |\hat{a}| \sin(\alpha - \beta) \underline{n} = \underline{k} (\sin\alpha \cos\beta - \cos\alpha \sin\beta)$$

$$\sin(\alpha - \beta) \underline{k} = \underline{k} (\sin\alpha \cos\beta - \cos\alpha \sin\beta)$$



$$(\therefore \hat{|\underline{b}|} = 1, \hat{|\underline{a}|} = 1)$$

$$\sin(\alpha - \beta) = \sin\alpha \cos\beta - \cos\alpha \sin\beta \quad \text{Hence proved}$$

Q.9 If $\underline{a} \times \underline{b} = \underline{0}$ and $\underline{a} \cdot \underline{b} = 0$. What conclusion can be drawn about $\underline{a}$ or $\underline{b}$?
(Gujranwala Board 2004, 2007, Lahore Board 2009 (Supply))

Solution:

$$\text{If } \underline{a} \times \underline{b} = \underline{0} \quad \Rightarrow \text{(i) } \underline{a} \text{ and } \underline{b} \text{ are parallel} \quad \text{(ii) Either } \underline{a} = \underline{0} \text{ or } \underline{b} = \underline{0}$$

$$\text{If } \underline{a} \cdot \underline{b} = 0 \quad \Rightarrow \text{(iii) } \underline{a} \text{ and } \underline{b} \text{ are perpendicular} \quad \text{(iv) Either } \underline{a} = \underline{0} \text{ or } \underline{b} = \underline{0}$$

This is not possible that $\underline{a}$ and $\underline{b}$ are parallel and perpendicular at the same time

So either $\underline{a} = \underline{0}$ or $\underline{b} = \underline{0}$

$\therefore$ $\underline{a}$ and $\underline{b}$ are null vectors.

EXERCISE 7.5

Q.1 Find the volume of parallelepiped for which the given vectors are three edges.

(i) $\underline{u} = 3\underline{i} + 0\underline{j} + 2\underline{k}$; $\underline{v} = \underline{i} + 2\underline{j} + \underline{k}$; $\underline{w} = 0\underline{i} - \underline{j} + 4\underline{k}$

Solution:

Formula

$$\text{Volume of parallelepiped} = \underline{u} \cdot (\underline{v} \times \underline{w})$$

$$\begin{aligned} \underline{u} \cdot (\underline{v} \times \underline{w}) &= \begin{vmatrix} 3 & 0 & 2 \\ 1 & 2 & 1 \\ 0 & -1 & 4 \end{vmatrix} \\ &= 3 \begin{vmatrix} 2 & 1 \\ -1 & 4 \end{vmatrix} - 0 \begin{vmatrix} 1 & 1 \\ 0 & 4 \end{vmatrix} + 2 \begin{vmatrix} 1 & 2 \\ 0 & -1 \end{vmatrix} \\ &= 3(8 + 1) - 0 + 2(-1) = 27 - 2 = 25 \text{ cubic units} \quad \text{Ans.} \end{aligned}$$

(ii) $\underline{u} = \underline{i} - 4\underline{j} - \underline{k}$; $\underline{v} = \underline{i} - \underline{j} - 2\underline{k}$; $\underline{w} = 2\underline{i} - 3\underline{j} + \underline{k}$

Solution:

$$\text{Volume of parallelepiped} = \underline{u} \cdot (\underline{v} \times \underline{w})$$

$$= \begin{vmatrix} 1 & -4 & -1 \\ 1 & -1 & -2 \\ 2 & -3 & 1 \end{vmatrix}$$



$$\begin{aligned}
 &= 1 \begin{vmatrix} -1 & -2 \\ -3 & 1 \end{vmatrix} + 4 \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} - 1 \begin{vmatrix} 1 & -1 \\ 2 & -3 \end{vmatrix} \\
 &= 1(-1-6) + 4(1+4) - 1(-3+2) = -7 + 20 + 1 = 14 \text{ cubic units Ans.}
 \end{aligned}$$

(iii) $\underline{u} = \underline{i} - 2\underline{j} + 3\underline{k}$; $\underline{v} = 2\underline{i} - \underline{j} - \underline{k}$; $\underline{w} = \underline{j} + \underline{k}$

Solution:

Volume of parallelepiped $= \underline{u} \cdot (\underline{v} \times \underline{w})$

$$\begin{aligned}
 \underline{u} \cdot (\underline{v} \times \underline{w}) &= \begin{vmatrix} 1 & -2 & 3 \\ 2 & -1 & -1 \\ 0 & 1 & 1 \end{vmatrix} \\
 &= 1 \begin{vmatrix} -1 & -1 \\ 1 & 1 \end{vmatrix} + 2 \begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix} + 3 \begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix} \\
 &= 1(-1+1) + 2(2-0) + 3(2-0) \\
 &= 4 + 6 = 10 \text{ cubic units Ans.}
 \end{aligned}$$

Q.2 Verify that $\underline{a} \cdot \underline{b} \times \underline{c} = \underline{b} \cdot \underline{c} \times \underline{a} = \underline{c} \cdot \underline{a} \times \underline{b}$

If $\underline{a} = 3\underline{i} - \underline{j} + 5\underline{k}$, $\underline{b} = 4\underline{i} + 3\underline{j} - 2\underline{k}$, $\underline{c} = 2\underline{i} + 5\underline{j} + \underline{k}$

(Gujranwala Board, 2003, Lahore Board 2007)

Solution:

$$\begin{aligned}
 \underline{a} \cdot (\underline{b} \times \underline{c}) &= \begin{vmatrix} 3 & -1 & 5 \\ 4 & 3 & -2 \\ 2 & 5 & 1 \end{vmatrix} \\
 &= 3(3+10) + 1(4+4) + 5(20-6) \\
 &= 39 + 8 + 70 = 117 \quad \text{..... (i)}
 \end{aligned}$$

$$\begin{aligned}
 \underline{b} \cdot (\underline{c} \times \underline{a}) &= \begin{vmatrix} 4 & 3 & -2 \\ 2 & 5 & 1 \\ 3 & -1 & 5 \end{vmatrix} \\
 &= 4(25+1) - 3(10-3) - 2(-2-15) \\
 &= 104 - 21 + 34 = 117 \quad \text{..... (ii)}
 \end{aligned}$$

$$\begin{aligned}
 \underline{c} \cdot (\underline{a} \times \underline{b}) &= \begin{vmatrix} 2 & 5 & 1 \\ 3 & -1 & 5 \\ 4 & 3 & -2 \end{vmatrix} \\
 &= 2(2-15) - 5(-6-20) + 1(9+4)
 \end{aligned}$$



$$= -26 + 130 + 13 = 117 \quad \dots\dots (iii)$$

From (i), (ii) & (iii) it is verified that

$$\underline{a} \cdot \underline{b} \times \underline{c} = \underline{b} \cdot \underline{c} \times \underline{a} = \underline{c} \cdot \underline{a} \times \underline{b}$$

Q.3 Prove that the vectors $\underline{i} - 2\underline{j} + 3\underline{k}$, $-2\underline{i} + 3\underline{j} - 4\underline{k}$ and $\underline{i} - 3\underline{j} + 5\underline{k}$ are coplanar. (Gujranwala Board 2007)

Solution:

$$\text{Let } \underline{u} = \underline{i} - 2\underline{j} + 3\underline{k}, \quad \underline{v} = -2\underline{i} + 3\underline{j} - 4\underline{k}, \quad \underline{w} = \underline{i} - 3\underline{j} + 5\underline{k}$$

$$\begin{aligned} \underline{u} \cdot \underline{v} \times \underline{w} &= \begin{vmatrix} 1 & -2 & 3 \\ -2 & 3 & -4 \\ 1 & -3 & 5 \end{vmatrix} \\ &= 1(15 - 12) + 2(-10 + 4) + 3(6 - 3) = 3 - 12 + 9 = 12 - 12 = 0 \\ \underline{u} \cdot \underline{v} \times \underline{w} &= 0 \end{aligned}$$

Hence $\underline{u}, \underline{v}, \underline{w}$ are coplanar.

Q.4 Find the constant α such that the vectors are coplanar.

(i) $\underline{i} - \underline{j} + \underline{k}$, $\underline{i} - 2\underline{j} - 3\underline{k}$ & $3\underline{i} - \alpha\underline{j} + 5\underline{k}$ (Lahore Board 2007, 2009)

Solution:

$$\begin{aligned} \text{Let } \underline{a} &= \underline{i} - \underline{j} + \underline{k} \\ \underline{b} &= \underline{i} - 2\underline{j} - 3\underline{k} \\ \underline{c} &= 3\underline{i} - \alpha\underline{j} + 5\underline{k} \end{aligned}$$

Since given vectors are coplanar so

$$\begin{aligned} \underline{a} \cdot \underline{b} \times \underline{c} &= 0 \\ \begin{vmatrix} 1 & -1 & 1 \\ 1 & -2 & -3 \\ 3 & -\alpha & 5 \end{vmatrix} &= 0 \\ 1(-10 - 3\alpha) + 1(5 + 9) + 1(-\alpha + 6) &= 0 \\ -10 - 3\alpha + 14 - \alpha + 6 &= 0 \\ -4\alpha + 10 &= 0 \\ 4\alpha &= 10 \\ \alpha &= \frac{10}{4} = \frac{5}{2} \quad \text{Ans.} \end{aligned}$$

(ii) $\underline{a} = \underline{i} - 2\alpha\underline{j} - \underline{k}$, $\underline{b} = \underline{i} - \underline{j} + 2\underline{k}$, $\underline{c} = \alpha\underline{i} - 2\underline{j} + \underline{k}$

Solution:

Since $\underline{a}, \underline{b}, \underline{c}$ are coplanar so $\underline{a} \cdot \underline{b} \times \underline{c} = 0$



$$\begin{vmatrix} 1 & -2\alpha & -1 \\ 1 & -1 & 2 \\ \alpha & -2 & 1 \end{vmatrix} = 0$$

$$1(-1+4) + 2\alpha(1-2\alpha) - 1(-2+\alpha) = 0$$

$$3 + 2\alpha - 4\alpha^2 + 2 - \alpha = 0$$

$$-4\alpha^2 + \alpha + 5 = 0$$

$$4\alpha^2 - \alpha - 5 = 0$$

$$4\alpha^2 - 5\alpha + 4\alpha - 5 = 0$$

$$\alpha(4\alpha - 5) + 1(4\alpha - 5) = 0$$

$$(4\alpha - 5)(\alpha + 1) = 0$$

$$4\alpha - 5 = 0, \quad \alpha + 1 = 0$$

$$4\alpha = 5, \quad \alpha = -1$$

$$\alpha = \frac{5}{4}, \quad \alpha = -1 \quad \text{Ans.}$$

Q.5 Find the value of

(i) $2\mathbf{i} \times 2\mathbf{j} \cdot \mathbf{k}$

Solution:

$$2\mathbf{i} \times 2\mathbf{j} \cdot \mathbf{k} = 4(\mathbf{i} \times \mathbf{j}) \cdot \mathbf{k} = 4(\mathbf{k} \cdot \mathbf{k}) = 4$$

(ii) $3\mathbf{j} \cdot \mathbf{k} \times \mathbf{i}$

Solution:

$$3\mathbf{j} \cdot \mathbf{k} \times \mathbf{i} = 3\mathbf{j} \cdot \mathbf{j} = 3(1) = 3 \quad \text{Ans.}$$

(iii) $[\mathbf{k} \ \mathbf{i} \ \mathbf{j}]$

Solution:

$$= \frac{\mathbf{k} \cdot \mathbf{i} \times \mathbf{j}}{[\mathbf{i} \ \mathbf{i} \ \mathbf{k}]} = \frac{\mathbf{k} \cdot \mathbf{k}}{[\mathbf{i} \ \mathbf{i} \ \mathbf{k}]} = 1 \quad \text{Ans.}$$

(iv) $[\mathbf{i} \ \mathbf{i} \ \mathbf{k}]$

Solution:

$$= \frac{\mathbf{i} \cdot \mathbf{i} \times \mathbf{k}}{[\mathbf{i} \ \mathbf{i} \ \mathbf{k}]} = \frac{\mathbf{i} \cdot (-\mathbf{j})}{[\mathbf{i} \ \mathbf{i} \ \mathbf{k}]} = -(\mathbf{i} \cdot \mathbf{j}) = -0 = 0 \quad \text{Ans.}$$

Q.5(b) Prove that $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) + \mathbf{v} \cdot (\mathbf{w} \times \mathbf{u}) + \mathbf{w} \cdot (\mathbf{u} \times \mathbf{v}) = 3\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$

(Lahore Board, 2011)

Solution:

$$\text{L.H.S. } \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) + \mathbf{v} \cdot (\mathbf{w} \times \mathbf{u}) + \mathbf{w} \cdot (\mathbf{u} \times \mathbf{v})$$

We know that

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \mathbf{v} \cdot (\mathbf{u} \times \mathbf{w}) = \mathbf{w} \cdot (\mathbf{v} \times \mathbf{u})$$

Putting values in L.H.S.

$$= \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) + \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) + \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$$



$$= 3\vec{u} \cdot (\vec{v} \times \vec{w})$$

R.H.S Hence proved

Q.6 Find volume of tetrahedron with the vertices

(i) $(0, 1, 2)$, $(3, 2, 1)$, $(1, 2, 1)$ & $(5, 5, 6)$

Solution:

Formula

Volume of tetrahedron when A,B,C, D whose vertices are given $= \frac{1}{6} (\vec{AB} \cdot \vec{AC} \times \vec{AD})$

Let A $(0, 1, 2)$, B $(3, 2, 1)$. C $(1, 2, 1)$ D $(5, 5, 6)$

$$\begin{aligned} \vec{AB} &= \text{Position vector of B} - \text{Position vector of A} \\ &= (3-0)\underline{i} + (2-1)\underline{j} + (1-2)\underline{k} \\ &= 3\underline{i} + \underline{j} - \underline{k} \end{aligned}$$

$$\begin{aligned} \vec{AC} &= \text{Position vector of C} - \text{Position vector of A} \\ &= (1-0)\underline{i} + (2-1)\underline{j} + (1-2)\underline{k} \end{aligned}$$

$$\vec{AC} = \underline{i} + \underline{j} - \underline{k}$$

$$\begin{aligned} \vec{AD} &= \text{Position vector of D} - \text{Position vector of A} \\ &= (5-0)\underline{i} + (5-1)\underline{j} + (6-2)\underline{k} \end{aligned}$$

$$\vec{AD} = 5\underline{i} + 4\underline{j} + 4\underline{k}$$

Now

$$\begin{aligned} \vec{AB} \cdot \vec{AC} \times \vec{AD} &= \begin{vmatrix} 3 & 1 & -1 \\ 1 & 1 & -1 \\ 5 & 4 & 4 \end{vmatrix} \\ &= 3(4+4) - 1(4+5) - 1(4-5) = 24 - 9 + 1 = 16 \end{aligned}$$

Volume of tetrahedron $= \frac{1}{6} (16) = \frac{8}{3}$ cubic units Ans.

(ii) $(2, 1, 8)$, $(3, 2, 9)$, $(2, 1, 4)$ & $(3, 3, 10)$ (Lahore Board 2011)

Solution:

Let A $(2, 1, 8)$, B $(3, 2, 9)$, C $(2, 1, 4)$, D $(3, 3, 10)$

$$\begin{aligned} \vec{AB} &= \vec{OB} - \vec{OA} \\ &= (3-2)\underline{i} + (2-1)\underline{j} + (9-8)\underline{k} \end{aligned}$$

$$\vec{AB} = \underline{i} + \underline{j} + \underline{k}$$



$$\begin{aligned}\vec{AC} &= \vec{OC} - \vec{OA} \\ &= (2-2)\underline{i} + (1-1)\underline{j} + (4-8)\underline{k}\end{aligned}$$

$$\vec{AC} = 0\underline{i} + 0\underline{j} - 4\underline{k}$$

$$\begin{aligned}\vec{AD} &= \vec{OD} - \vec{OA} \\ &= (3-2)\underline{i} + (3-1)\underline{j} + (10-8)\underline{k}\end{aligned}$$

$$\vec{AD} = \underline{i} + 2\underline{j} + 2\underline{k}$$

$$\begin{aligned}\vec{AB} \cdot \vec{AC} \times \vec{AD} &= \begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & -4 \\ 1 & 2 & 2 \end{vmatrix} \\ &= 1(0+8) - 1(0+4) + 1(0-0) = 8 - 4 = 4\end{aligned}$$

$$\text{Volume of tetrahedron} = \frac{1}{6} (\vec{AB} \cdot \vec{AC} \times \vec{AD}) = \frac{1}{6} (4) = \frac{2}{3} \text{ cubic units} \quad \text{Ans.}$$

Q.7 Find the work done, if the point at which the constant force $\vec{F} = 4\underline{i} + 3\underline{j} + 5\underline{k}$ is applied to an object, moves from $P_1(3, 1, -2)$ to $P_2(2, 4, 6)$
(Gujranwala Board, 2004)

Solution:

$$\text{Given } \vec{F} = 4\underline{i} + 3\underline{j} + 5\underline{k}$$

$$\begin{aligned}\vec{d} = \vec{P_1P_2} &= \vec{OP_2} - \vec{OP_1} \\ &= (2-3)\underline{i} + (4-1)\underline{j} + (6+2)\underline{k}\end{aligned}$$

$$\vec{d} = -\underline{i} + 3\underline{j} + 8\underline{k}$$

$$\begin{aligned}\text{Work done} &= \vec{F} \cdot \vec{d} \\ &= (4\underline{i} + 3\underline{j} + 5\underline{k}) \cdot (-\underline{i} + 3\underline{j} + 8\underline{k}) \\ &= -4 + 9 + 40 \\ &= 45 \quad \text{Ans.}\end{aligned}$$

Q.8 A particle, acted by constant forces $4\underline{i} + \underline{j} - 3\underline{k}$ and $3\underline{i} - \underline{j} - \underline{k}$ is displacement from $A(1, 2, 3)$ to $B(5, 4, 1)$. Find the work done.

Solution:

$$\vec{F_1} = 4\underline{i} + \underline{j} - 3\underline{k}, \quad \vec{F_2} = 3\underline{i} - \underline{j} - \underline{k}$$



$$\begin{aligned}
 \vec{F} &= \vec{F}_1 + \vec{F}_2 = 7\vec{i} + 0\vec{j} - 4\vec{k}, \\
 \vec{d} = \vec{AB} &= \vec{OB} - \vec{OA} \\
 &= (5-1)\vec{i} + (4-2)\vec{j} + (1-3)\vec{k} \\
 \vec{d} &= 4\vec{i} + 2\vec{j} - 2\vec{k} \\
 \text{Work done} &= \vec{F} \cdot \vec{d} \\
 &= (7\vec{i} + 0\vec{j} - 4\vec{k}) \cdot (4\vec{i} + 2\vec{j} - 2\vec{k}) \\
 &= 28 + 0 + 8 \\
 &= 36 \quad \text{Ans.}
 \end{aligned}$$

Q.9 A particle is displaced from the point A(5, -5, -7) to the point B(6, 2, -2) under the action of constant forces defined by $10\vec{i} - \vec{j} + 11\vec{k}$, $4\vec{i} + 5\vec{j} + 9\vec{k}$ and $-2\vec{i} + \vec{j} - 9\vec{k}$. Show that the total work done by the forces is 102 unit.

Solution:

$$\begin{aligned}
 \vec{d} = \vec{AB} &= \vec{OB} - \vec{OA} \\
 &= (6-5)\vec{i} + (2+5)\vec{j} + (-2+7)\vec{k} \\
 \vec{d} &= \vec{i} + 7\vec{j} + 5\vec{k} \\
 \vec{F}_1 &= 10\vec{i} - \vec{j} + 11\vec{k}, \quad \vec{F}_2 = 4\vec{i} + 5\vec{j} + 9\vec{k}, \quad \vec{F}_3 = 2\vec{i} + \vec{j} - 9\vec{k} \\
 \text{Total forces } \vec{F} &= \vec{F}_1 + \vec{F}_2 + \vec{F}_3 \\
 &= 10\vec{i} - \vec{j} + 11\vec{k} + 4\vec{i} + 5\vec{j} + 9\vec{k} + 2\vec{i} + \vec{j} - 9\vec{k} \\
 \vec{F} &= 12\vec{i} + 5\vec{j} + 11\vec{k} \\
 \text{Work done} &= \vec{F} \cdot \vec{d} \\
 &= (12\vec{i} + 5\vec{j} + 11\vec{k}) \cdot (\vec{i} + 7\vec{j} + 5\vec{k}) \\
 &= 12 + 35 + 55 \\
 &= 102 \text{ units} \quad \text{Hence proved} \quad \text{Ans.}
 \end{aligned}$$

Q.10 A force of magnitude 6 units acting parallel to $2\vec{i} - 2\vec{j} + \vec{k}$ displaces, the point of application from (1, 2, 3) to (5, 3, 7). Find the work done.

Solution:

$$\begin{aligned}
 \text{Let } \vec{u} &= 2\vec{i} - 2\vec{j} + \vec{k} \\
 |\vec{u}| &= \sqrt{(2)^2 + (-2)^2 + (1)^2} = \sqrt{4+4+1} = \sqrt{9} = 3
 \end{aligned}$$



$$\hat{n} = \frac{\underline{u}}{|\underline{u}|} = \frac{2\underline{i} - 2\underline{j} + \underline{k}}{\sqrt{3}}$$

Hence, the force of magnitude 6 units is

$$\begin{aligned}\vec{F} &= 6\hat{n} \\ &= 6\left(\frac{2\underline{i} - 2\underline{j} + \underline{k}}{3}\right)\end{aligned}$$

$$\vec{F} = 4\underline{i} - 4\underline{j} + 2\underline{k}$$

Let

$$A(1, 2, 3), \quad B(5, 3, 7)$$

$$\begin{aligned}\vec{d} = \vec{AB} &= \vec{OB} - \vec{OA} \\ &= (5-1)\underline{i} + (3-2)\underline{j} + (7-3)\underline{k}\end{aligned}$$

$$\vec{d} = 4\underline{i} + \underline{j} + 4\underline{k}$$

$$\begin{aligned}\text{Work done} &= \vec{F} \cdot \vec{d} \\ &= (4\underline{i} - 4\underline{j} + 2\underline{k}) \cdot (4\underline{i} + \underline{j} + 4\underline{k}) \\ &= 16 - 4 + 8 = 20 \quad \text{Ans.}\end{aligned}$$

Q.11 A force $\vec{F} = 3\underline{i} + 2\underline{j} - 4\underline{k}$ is applied at the point $(1, -1, 2)$. Find the moment of the force about the point $(2, -1, 3)$

Solution:

$$\vec{F} = 3\underline{i} + 2\underline{j} - 4\underline{k}$$

$$\text{Let } A(1, -1, 2) \text{ and } B(2, -1, 3)$$

$$\begin{aligned}\vec{r} = \vec{BA} &= \vec{OA} - \vec{OB} \\ &= (1-2)\underline{i} + (-1+1)\underline{j} + (2-3)\underline{k}\end{aligned}$$

$$\vec{r} = -\underline{i} + 0\underline{j} - \underline{k}$$

$$\begin{aligned}\text{Required moment } M &= \vec{r} \times \vec{F} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -1 & 0 & -1 \\ 3 & 2 & -4 \end{vmatrix} \\ &= \underline{i} \begin{vmatrix} 0 & -1 \\ 2 & -4 \end{vmatrix} - \underline{j} \begin{vmatrix} -1 & -1 \\ 3 & -4 \end{vmatrix} + \underline{k} \begin{vmatrix} -1 & 0 \\ 3 & 2 \end{vmatrix} \\ &= \underline{i} (0+2) - \underline{j} (4+3) + \underline{k} (-2-0)\end{aligned}$$



$$= 2\underline{i} - 7\underline{j} - 2\underline{k} \quad \text{Ans.}$$

Q.12 A force $\vec{F} = 4\underline{i} - 3\underline{k}$ passes through the point A(2, -2, 5). Find the moment of $\vec{F}$ about point B (1, -3, 1) (Lahore Board 2009)

Solution:

$$\begin{aligned} \vec{F} &= 4\underline{i} + 0\underline{j} - 3\underline{k} \\ \vec{r} = \vec{BA} &= \vec{OA} - \vec{OB} \\ &= (2-1)\underline{i} + (-2+3)\underline{j} + (5-1)\underline{k} \\ \vec{r} &= \underline{i} + \underline{j} + 4\underline{k} \end{aligned}$$

Required moment

$$\begin{aligned} \vec{r} \times \vec{F} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 1 & 4 \\ 4 & 0 & -3 \end{vmatrix} = \underline{i} \begin{vmatrix} 1 & 4 \\ 0 & -3 \end{vmatrix} - \underline{j} \begin{vmatrix} 1 & 4 \\ 4 & -3 \end{vmatrix} + \underline{k} \begin{vmatrix} 1 & 1 \\ 4 & 0 \end{vmatrix} \\ &= \underline{i}(-3-0) - \underline{j}(-3-16) + \underline{k}(0-4) \\ \vec{r} \times \vec{F} &= -3\underline{i} + 19\underline{j} - 4\underline{k} \quad \text{Ans.} \end{aligned}$$

Q.13 Give a force $\vec{F} = 2\underline{i} + \underline{j} - 3\underline{k}$ acting at a point A (1, -2, 1). Find the moment of $\vec{F}$ about the point B (2, 0, -2)

Solution:

$$\begin{aligned} \vec{F} &= 2\underline{i} + \underline{j} - 3\underline{k} \\ \vec{r} = \vec{BA} &= \vec{OA} - \vec{OB} \\ &= (1-2)\underline{i} + (-2-0)\underline{j} + (1+2)\underline{k} \\ \vec{r} &= -\underline{i} - 2\underline{j} + 3\underline{k} \end{aligned}$$

Required moment

$$\vec{r} \times \vec{F} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -1 & -2 & 3 \\ 2 & 1 & -3 \end{vmatrix}$$



$$\begin{aligned}
 &= \underline{i} \begin{vmatrix} -2 & 3 \\ 1 & -3 \end{vmatrix} - \underline{j} \begin{vmatrix} -1 & 3 \\ 2 & -3 \end{vmatrix} + \underline{k} \begin{vmatrix} -1 & -2 \\ 2 & 1 \end{vmatrix} \\
 &= \underline{i} (6 - 3) - \underline{j} (3 - 6) + \underline{k} (-1 + 4) \\
 &= 3\underline{i} + 3\underline{j} + 3\underline{k} \quad \text{Ans.}
 \end{aligned}$$

Q.14 Find the moment about A(1, 1, 1) of each of the concurrent forces $\underline{i} - 2\underline{j}$, $3\underline{i} + 2\underline{j} - \underline{k}$, $5\underline{j} + 2\underline{k}$, where P (2, 0, 1) is their point of concurrency.

(Lahore Board 2009)

Solution:

$$\begin{aligned}
 \vec{r} = \vec{AP} &= \vec{OP} - \vec{OA} \\
 &= (2 - 1)\underline{i} + (0 - 1)\underline{j} + (1 - 1)\underline{k} \\
 \vec{r} &= \underline{i} - \underline{j} + 0\underline{k} \\
 \vec{F}_1 &= \underline{i} - 2\underline{j}, \quad \vec{F}_2 = 3\underline{i} + 2\underline{j} - \underline{k}, \quad \vec{F}_3 = 0\underline{i} + 5\underline{j} + 2\underline{k} \\
 \vec{F} &= \vec{F}_1 + \vec{F}_2 + \vec{F}_3 \\
 &= \underline{i} - 2\underline{j} + 3\underline{i} + 2\underline{j} - \underline{k} + 0\underline{i} + 5\underline{j} + 2\underline{k} \\
 \vec{F} &= 4\underline{i} + 5\underline{j} + \underline{k}
 \end{aligned}$$

Moment of force

$$\begin{aligned}
 \vec{r} \times \vec{F} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & -1 & 0 \\ 4 & 5 & 1 \end{vmatrix} \\
 &= \underline{i} \begin{vmatrix} -1 & 0 \\ 5 & 1 \end{vmatrix} - \underline{j} \begin{vmatrix} 1 & 0 \\ 4 & 1 \end{vmatrix} + \underline{k} \begin{vmatrix} 1 & -1 \\ 4 & 5 \end{vmatrix} \\
 &= \underline{i} (-1 - 0) - \underline{j} (1 - 0) + \underline{k} (5 + 4) \\
 \vec{r} \times \vec{F} &= -\underline{i} - \underline{j} + 9\underline{k} \quad \text{Ans.}
 \end{aligned}$$

Q.15 A force $\vec{F} = 7\underline{i} + 4\underline{j} - 3\underline{k}$ is applied at P (1, -2, 3). Find its moment about the point Q (2, 1, 1).

Solution:

$$\begin{aligned}
 \vec{F} &= 7\underline{i} + 4\underline{j} - 3\underline{k} \\
 \vec{r} = \vec{QP} &= \vec{OD} - \vec{OQ}
 \end{aligned}$$



$$= (1 - 2)\underline{i} + (-2 - 1)\underline{j} + (3 - 1)\underline{k}$$

$$\vec{r} = -\underline{i} - 3\underline{j} + 2\underline{k}$$

Moment of force

$$\begin{aligned}\vec{r} \times \vec{F} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -1 & -3 & 2 \\ 7 & 4 & -3 \end{vmatrix} \\ &= \underline{i} \begin{vmatrix} -3 & 2 \\ 4 & -3 \end{vmatrix} - \underline{j} \begin{vmatrix} -1 & 2 \\ 7 & -3 \end{vmatrix} + \underline{k} \begin{vmatrix} -1 & -3 \\ 7 & 4 \end{vmatrix} \\ &= \underline{i} (9 - 8) - \underline{j} (3 - 14) + \underline{k} (-4 + 21)\end{aligned}$$

$$\vec{r} \times \vec{F} = \underline{i} + 11\underline{j} + 17\underline{k} \quad \text{Ans.}$$

